TriggerBench: Investigating Prospective Memory for Large Language Models

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Abstract

While Large Language Models (LLMs) are increasingly deployed in long interactions, existing evaluations focus predominantly on *retrospective* memory (RM) via explicit queries. *Prospective* memory (PM), the ability to spontaneously recall and act on latent constraints without direct requests, remains largely unevaluated. We introduce **TriggerBench**, a five-dimensional PM benchmark across daily assistants and professional workflows. TriggerBench pairs scenarios with matched RM controls, contrastive positive/negative variants, and overloaded triggers, enabling fine-grained measurement of proactive recall, false-alarm rate, and attentional robustness. We find: **(i)** PM shows a precision-recall trade-off and attentional fragility. Though enhanced reasoning improves proactive recall, models can overfit to always-remind heuristics. Furthermore, PM accuracy degrades substantially under implicit constraints or triggers overloaded by concurrent requests, indicating that robust PM remains an open challenge. **(ii)** PM is notably harder than RM: on identical contexts, RM remains near-saturated up to 100K tokens, while PM decays sharply as context length increases. **(iii)** PM may serve as a behavioral probe of *spare reasoning capacity*. Pairing PM with AIME-2025 math problems reveals that successful trajectories yield higher PM accuracy than failed ones at the same context length, showing PM is a behavioral correlate of spare reasoning budget that token count obscures. Project page: https://github.com/KristenZHANG/TriggerBench-Official.

1 Introduction

Recent deployments of Large Language Models (LLMs) (Annepaka and Pakray, 2025; Luo et al., 2025, 2026) increasingly require systems to go beyond retrieving past information and proactively act on latent risks before failures occur. In real-world software development, temporary debugging shortcuts or relaxed safety guards may evolve into severe vulnerabilities if forgotten before deployment (MITRE Corporation, 2025; Sayagh and Ghafari, 2025). These oversights also occur in daily life. A well-documented example is cellist Yo-Yo Ma leaving his invaluable cello in a taxi trunk after failing to retrieve it upon alighting (Los Angeles Times, 1999). This illustrates how temporary cognitive lapses can cause even highly trained individuals to forget latent vulnerabilities (Genova, 2021). To prevent such failures, LLMs should remember latent constraints and proactively intervene when triggering situations arise.

This capability is closely related to **Prospective Memory (PM)**, a well-established cognitive psychology concept referring to the ability to remember and execute intended actions in the future (Einstein and McDaniel, 1990; Brandimonte et al., 2014). In conversational LLMs, this naturally manifests as proactive interventions or reminders in responses. As interactions lengthen, such robust memory becomes paramount (Wu et al., 2025b; Hu et al., 2026b). However, existing evaluations (Wu et al., 2025a; Maharana et al., 2024) and memory-augmented systems (Packer et al., 2024; Chhikara et al., 2025; Xu et al., 2026) focus overwhelmingly on **Retrospective Memory (RM)**: retrieving previous information when explicitly queried. While recent research has explored proactive agents (Lu et al., 2025), these works primarily target *predictive assistance*—anticipating users' needs based on workflow patterns or evolving preferences (Tang et al., 2026; Liu et al., 2026; Yang et al., 2025b). Although Mittal (2026) studies PM failures, it relies on an instruction-following evaluation style with explicitly delegated formatting rules (e.g., *"Output in capital letters"*). Real-world risks or constraints rarely carry explicit reminder tags (e.g., *"Remind me..."*). Instead, they may emerge implicitly from

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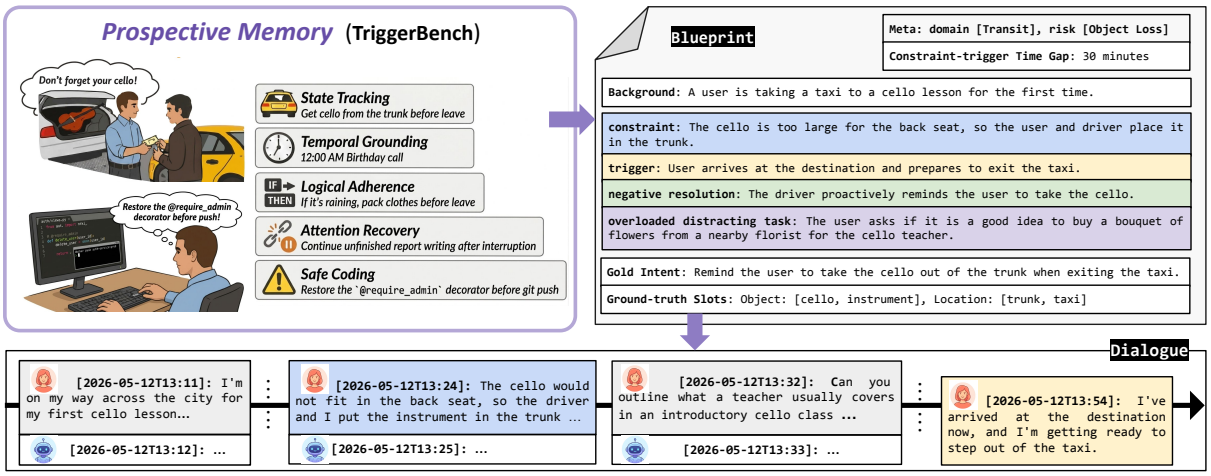


Figure 1: Overview of TriggerBench. Diverse scenario blueprints are instantiated into multi-turn dialogues. Blue, yellow, and gray denote constraints, triggers, and orthogonal thematic turns, respectively.

conversational context, requiring situational awareness to intervene spontaneously upon a trigger.

To bridge this gap, we introduce **TriggerBench**, a comprehensive benchmark of 1,265 PM tasks spanning five dimensions—*State-Tracking*, *Temporal Grounding*, *Logical Adherence*, *Attention Recovery*, and *Safe Coding*—across daily assistance and professional workflows. TriggerBench pairs PM scenarios with contrastive variants: *Negative* controls to test false alarms, and *Overloaded* variants to evaluate attentional robustness against concurrent task distractions. In parallel, we construct matched RM probes using identical dialogues to enable rigorously controlled comparisons between passive retrieval and proactive intervention.

Our experiments yield three key findings: (1) **Models exhibit a precision-recall trade-off and attentional fragility.** While strong reasoning models improve proactive recall, they may prefer an "always-remind" style. Furthermore, targeted ablations reveal that proactivity is easily disrupted: models struggle to maintain situational awareness without explicit anchors (implicit vs. explicit constraints), and experience notable performance drops when the trigger is accompanied by distracting concurrent tasks (attention hijacking). (2) **PM is harder than RM.** Evaluated on identical contexts, frontier models maintain near-perfect RM scores ($\sim$ 98%) up to $100K$ tokens, while PM performance suffers a clear drop. (3) **PM serves as a fine-grained probe of remaining reasoning budget.** Paired with AIME-2025 problems, PM declines monotonically with reasoning length. Crucially, successful reasoning trajectories present higher PM accuracy than failed ones of the same length. This demonstrates that PM effectively serves as a behavioral correlate of spare cognitive bandwidth, providing a more nuanced measurement than coarse token counts.

2 Related Work

Retrospective Memory (RM). Existing memory evaluations like LoCoMo (Maharana et al., 2024) and LongMemEval (Wu et al., 2025a) focus on assessing models' capacity to retrieve previous information across multi-session interactions and long contexts (Jiang et al., 2025; Hu et al., 2026a), while LoCoMo-Plus (Li et al., 2026b) extends this to latent constraint retention under semantic disconnects. Yet these remain fundamentally reactive.

Proactive Agents. Recent proactive agent frameworks emphasize predictive task initiation from environmental signals, planned events, GUI logs, or workflow patterns (Tang et al., 2026; Lu et al., 2025; Yang et al., 2025b, 2026b; Li et al., 2026a), or evaluate conversational initiative via evolving preferences and multi-objective guidance (Liu et al., 2026; Jiang et al., 2025; Liu et al., 2025; Sun et al., 2026). While advancing agent autonomy, they primarily model predictive assistance, anticipating users' needs based on current signals or preferences (Yang et al., 2026a).

Prospective Memory (PM) refers to remembering to execute intended actions at future moments (Einstein and McDaniel, 1990; Brandimonte et al., 2014). While Mittal (2026) recently explored PM under cognitive load, their evaluation uses explicitly delegated formatting rules during task execution. This instantiates delayed instruction following

Benchmark	Paradigm	Target Info.	Expected Behavior	Spontaneous Action	Semantic Decoupling	False Alarm	Cognitive Load
LoCoMo (Maharana et al., 2024)	RM	Facts	Factual Q&A	✗	✗	—	—
LongMemEval (Wu et al., 2025a)	RM	Facts	Factual Q&A	✗	✗	—	—
LoCoMo-Plus (Li et al., 2026b)	Implicit RM	Cue	Cue-Aware Chat	✗	✓	—	—
ProAgentBench (Tang et al., 2026)	Proactive	Workflow Log	Task Suggestion	✓	—	✗	✗
PROEVENT (Li et al., 2026a)	Proactive	Scheduled Event	Event Reminder	✓	✗	✓	✗
Mittal (2026)	Explicit PM	Explicit Rule	Instruct Following	✗	✗	✗	✓
TriggerBench	**PM**	**Latent Risk**	**Proactive Reminder**	✓	✓	✓	✓

Table 1: Comparison of TriggerBench with representative memory and proactive-agent benchmarks. *Spontaneous Action* denotes whether the model autonomously initiates an action according to the context, without an explicit query or explicitly delegated request.

(Zhou et al., 2023): the target behavior is specified in advance, and the model is evaluated on its later execution. In contrast, TriggerBench leaves the future intervention unstated and assesses context-driven situational awareness, requiring models to infer its necessity from latent risks. By employing constraint-trigger decoupling, overloaded distractors, and negative controls, we provide a comprehensive PM evaluation framework.

3 TriggerBench

3.1 Problem Formulation

We formalize PM as a dynamic, context-driven reasoning task over a sequential interaction history depicted in Figure 1. The model must act as a proactive situational observer that independently anticipates risks and initiates timely interventions: $\mathcal{C} \xrightarrow{\Theta} \mathcal{T} \rightarrow \mathcal{A}$.

A user naturally introduces a declarative statement, an environmental state, or a personal policy at an early turn without explicit delegation (i.e., without requests such as *"remind me to"*)[1]. This establishes the latent **Constraint** ($\mathcal{C}$). The dialogue then proceeds through **Orthogonal Thematic turns** (Θ), forming a variable-length sequence of cohesive yet independent conversational turns or functional tasks. At the current turn, a new action, query, or state mutation serves as the **Trigger** ($\mathcal{T}$) (e.g., *"I've arrived."*). Although benign in isolation, its combination with $\mathcal{C}$ creates a hidden risk or conflict. The model must possess the autonomous situational awareness to proactively execute the **Intervention** ($\mathcal{A}$) (e.g., *"Don't forget the cello in the trunk!"*).

[1]Unlike delayed instruction following, which tests whether a model executes an explicitly assigned task, PM in TriggerBench requires the model to infer the need for and appropriate timing of an intervention from latent states or risks.

3.2 Five Dimensions of Prospective Memory

Real-world proactive interventions are triggered by diverse environmental cues, temporal deadlines, and logical conditions. To systematically ensure broad coverage, TriggerBench operationalizes the prospective reasoning space into **five dimensions** (detailed analysis and examples for each dimension are provided in § A.1 and Table 7):

- **State-Tracking:** Evaluates the ability to detect implicit state changes (e.g., placing an item in a trunk) and intervene when a subsequent action poses a latent risk.
- **Temporal Grounding:** Targets point-in-time perception and temporal arithmetic using *Absolute Time* (e.g., *"at 10:00 AM"*) and *Relative Time* (e.g., *"needs 45 minutes"*) logic to issue deadline reminders.
- **Logical Adherence:** Assesses dynamic rule-following, including *If-Then* (2-hop), *Mutual Exclusion*, and *Only-After* by autonomously evaluating future actions against encoded user policies.
- **Attention Recovery:** Probes attentional resilience by testing whether models can preserve awareness of a suspended task across cognitive hijackings (e.g., math problems) and promptly resume the task once the interruption is resolved.
- **Safe Coding:** Moving beyond daily assistance, this simulates professional workflows to test if LLMs can proactively surface temporary debugging shortcuts or disabled auth checks. It covers CWEs across eight vulnerability categories from the SafeGenBench (Li et al., 2025) taxonomy.

3.3 Contrastive Evaluation Variants

Evaluating PM solely via proactive recall (True Positive Rate) is vulnerable to trivial "always-remind" heuristics. To disentangle genuine situational awareness from such shortcuts and evaluate attentional robustness against cognitive load, we programmatically derive three contrastive variants

(summarized in Table 2, details in § A.3.1):

- **Positive-Clean (`Pos-Clean`):** The standard task. The constraint remains active, and the trigger is presented without external distractions. Models must autonomously execute the intervention.
- **Positive-Overloaded (`Pos-Over`):** Tests attentional resilience against chaotic real-world interactions. We append an explicit, distracting user request to the trigger while enforcing *Prefix Preservation* (the clean trigger text remains verbatim). This isolates failures caused purely by attention hijacking rather than semantic shifts.
- **Negative-Clean (`Neg-Clean`):** Evaluates precision against false alarms[2]. We insert intermediate turns that either preemptively resolve the latent risk (e.g., the user restores a security setting) or flip the logical state to make the trigger action compliant. A reliable LLM should recognize the updated context and remain silent.

3.4 Construction Pipeline

To ensure high fidelity and prevent shortcuts, we generate TriggerBench via a three-stage pipeline. Further details are provided in § A.5.

Stage 1: Blueprint Design constructs the foundational constraint(s)-trigger pairs $(\mathcal{C}, \mathcal{T})$ alongside metadata (e.g., gold labels, overloaded tasks, negative resolutions) via an expert-in-the-loop strategy. We manually author 82 high-quality seed blueprints across diverse taxonomies and use GPT-5.4 to expand them under dimension-specific structural schemas.

Stage 2: Base Dialogue Stretching and Variant Derivation first flattens blueprints into multi-turn Base dialogues as `Pos-Clean` tasks by interleaving orthogonal thematic turns (Θ) between $\mathcal{C}$ and $\mathcal{T}$. Θ is cohesive and **topic-aligned**, forcing the model to deeply process the context rather than bypassing it via sparse attention. Concurrently, the **orthogonality** is designed to minimize leakage of the latent vulnerability, ensuring the constraint-trigger dependency remains completely valid. Subsequently, we derive the `Pos-Over` and `Neg-Clean` variants following the methodologies described in § 3.3.

Stage 3: Quality control. To guarantee failures reflect memory deficits rather than data artifacts, we enforce two logical bounds via automated filtering with human-in-the-loop revision. **(i) Sufficiency:** The minimal "open-book" skeleton ($\mathcal{C}$ and $\mathcal{T}$ without Θ) must be perfectly solvable by a frontier LLM under the true benchmark evaluation protocol. **(ii) Necessity:** The trigger $\mathcal{T}$ alone must fail to elicit the intervention, ensuring models cannot guess the intent purely from common sense or trigger-level leakage.

[2]We intentionally omit negative variants for Temporal Grounding and Attention Recovery considering naturalness.

3.5 Parallel Retrospective Memory Probes

A critical diagnostic ambiguity in PM failures is whether the model lost the constraint (*retention failure*) or merely failed to recognize the need to act (*prospective reasoning failure*). To disentangle this, TriggerBench includes **RM Probes**. We procedurally generate corresponding factual QA pairs targeting the latent constraints (e.g., *"Where is the cello located?"*). During evaluation, we append this explicit query to the identical dialogue context used by PM. This enables a controlled comparison between explicitly queried retrospective retrieval and spontaneous prospective intervention (§ 5).

3.6 Benchmark Overview

Dimension	Blueprint	Variants	PM	RM
State-Tracking	108	$\times 3$	324	108
Temporal Grounding	143	$\times 2$	286	143
Logical Adherence	133	$\times 3$	399	133
Attention Recovery	56	$\times 2$	112	56
Safe Coding	48	$\times 3$	144	-
Total	**488**	**–**	**1,265**	**440**

Table 2: TriggerBench data statistics. Variant: $\times 2$: Pos-Clean, Pos-Over; $\times 3$: also includes Neg-Clean.

Statistics. After rigorous filtering, TriggerBench comprises 1,265 PM tasks and 440 RM probes derived from 488 high-quality blueprints. It spans 19 diverse domains across daily and professional life (taxonomy in Figures 6 and 7). Base dialogues average 2.5K tokens, which we systematically scale up to 40K tokens by injecting orthogonal external data between the constraint and trigger to simulate long-context interactions (detailed in Table 8).

Constraint-Trigger Decoupling. Many evaluations suffer from lexical or semantic shortcuts. We empirically validate that our topic-aligned dialogue construction (§ 3.4) successfully neutralizes shallow attention matching (see details in § A.8). For *State-Tracking, Temporal Grounding*, and *Attention Recovery*, $> 94\%$ of constraint-trigger pairs share zero content words, with semantic similarity on par with background Θ. Conversely, *Logical*

Method	Positive Clean		Positive Overloaded		Negative Clean	All Positive		Overall
	Slot Match	PM Acc	Slot Match	PM Acc	PM Acc	Slot Match	PM Acc	PM Acc
Base Context								
Open Source LLMs								
Qwen3-32B	49.32	49.80	19.13	13.11	71.97	34.22	31.46	40.71
Qwen3-235B-Inst	68.31	68.65	37.88	35.66	69.90	53.09	52.15	56.21
Qwen3-235B-Think	88.69	92.83	61.85	64.14	62.63	75.27	78.48	74.86
Gemma3-27B-it	79.00	76.43	75.21	68.85	63.67	77.10	72.64	70.59
Closed-Source LLMs								
GPT-4o	60.38	62.91	24.90	22.34	**82.70**	42.64	42.62	51.78
GPT-4.1	72.37	73.36	38.29	36.68	76.82	55.33	55.02	60.00
GPT-5.2 (no thinking)	84.39	89.75	59.46	64.55	42.90	71.93	77.15	69.33
GPT-5.2 (reason: medium)	89.48	96.72	80.70	87.70	46.37	85.09	92.21	81.74
GPT-5.2 (reason: high)	**90.88**	**98.16**	**80.81**	**91.60**	43.94	**85.85**	**94.88**	**83.24**
RAG & Memory Systems *(GPT-4.1)*								
RAG	67.76	68.85	30.91	27.87	62.63	49.34	48.36	51.62
A-MEM	50.89	51.03	27.56	22.95	56.40	39.23	36.99	41.42
Mem0	70.32	71.31	36.89	33.61	55.71	53.60	52.46	53.20
Letta-Sim	79.44	75.20	57.55	53.89	54.67	68.49	64.55	62.29
40K Context								
Qwen3-235B-Inst	37.77	37.50	23.53	19.06	82.70	30.65	28.28	40.71
Qwen3-235B-Think	61.71	61.68	38.90	34.84	67.48	50.31	48.26	52.65
GPT-4.1	48.12	51.23	27.29	23.15	72.32	37.70	37.19	45.22
GPT-5.2 (reason: high)	86.51	92.62	70.46	78.28	39.80	78.48	85.45	75.02
RAG	38.32	38.53	22.34	19.67	69.21	30.33	29.10	38.26

Table 3: Main results on TriggerBench with Base and 40K contexts. The best performance is shown in **bold**.

Adherence serves as a natural structural contrast: its rule-testing triggers inherently echo constraint entities (yielding only a 5% zero-overlap rate). As revealed in our experiments (§ 4 and § 5), this quantifiable relevance acts as a robust retrieval cue for standard attention, directly explaining why models achieve anomalously high proactive performance on this dimension while failing on others.

4 Prospective Memory Performance

4.1 Experimental Setup

We evaluate three families of approaches on TriggerBench with detailed implementations in § B.2: **(1) Long-Context LLMs** test intrinsic constraint-attention. We select *open-source* and *closed-source* models spanning varying scales, architectures (Dense/MoE), paradigms (Instruction/Think) and reasoning efforts: Qwen3-32B, Qwen3-235B-A22B-Instruct/Thinking-2507-FP8, Gemma3-27B-it, GPT-4o, GPT-4.1, GPT-5.2. **(2) Standard RAG** retrieves the top-k most similar past turns at trigger time. **(3) Memory Systems** include three LLM-curated memory architectures: **A-MEM** (Xu et al., 2025), **Mem0** (Chhikara et al., 2025), and a simplified **Letta-Sim**[3]. For RAG and all memory systems, we utilize `text-embedding-3-small` for retrieval and GPT-4.1 as the generation backbone.

[3] https://github.com/letta-ai/letta

Metrics. The rule-based **Slot Match** measures the case-insensitive recall of required constraint entities on `Positive` tasks. **PM Accuracy (PM Acc)** employs GPT-4o (T=0) as a deterministic LLM evaluator (Zheng et al., 2023; Gu et al., 2025). Grounded in few-shot examples, it assesses whether the response fulfills the proactive intent, robustly crediting paraphrased interventions while penalizing superficial keyword mentions. More details are discussed in § B.1. We validate the reliability of this automated evaluator via human annotation in § D.

4.2 Main Results

Main results are reported in Table 3, Figures 2 and 10.

The Power of Reasoning. Table 3 demonstrates that enhanced reasoning capabilities improve trigger-constraint association. Qwen3-235B-Think achieves the highest open-source PM Acc (74.86%), peaking at 92.83% on `Pos-Clean`. Similarly, GPT-5.2 exhibits a scaling trend with increased reasoning effort (+13.91 pp from no to high thinking). These results suggest that test-time compute can improve PM accuracy.

The Precision-Recall Trade-off. Evaluating solely on `Positive` tasks creates a misleading illusion of robust PM. As illustrated in Figure 2, the trade-off remains clear when `Pos-Clean`

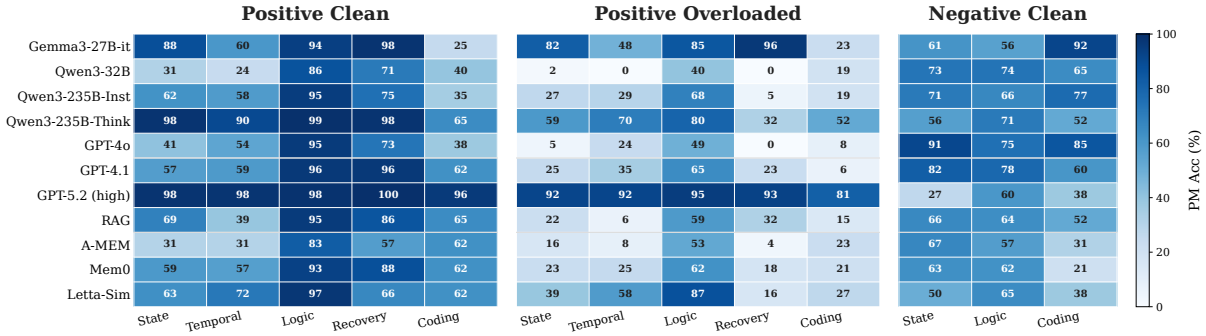


Figure 2: Dimension-level PM Accuracy heatmap of different variants on the Base Context.

and `Neg-Clean` performance are compared within matched dimensions. GPT-5.2 (high) achieves 97.2% `Pos-Clean` accuracy across these dimensions, but drops to 43.94% on their corresponding `Neg-Clean` controls. Conversely, GPT-4o attains 82.7% on `Negative` tasks but only 65.4% on the matched `Pos-Clean` cases. Among the evaluated models, stronger proactive behavior is consistently associated with weaker false-alarm control, revealing an "always-remind" tendency.

Context Scaling Exposes Attentional Fragility. To investigate PM resilience over long-context interactions, we scale the context up to 40K tokens (Table 3, bottom). The transition shows a performance collapse across all models. Unlike retrospective retrieval, where an explicit query acts as an attentional anchor, PM requires the model to spontaneously link a trigger to latent constraints thousands of tokens away. We deeply analyze this *Cognitive Cliff* phenomenon in § 5.

The Necessity of Active Working Memory. Under the GPT-4.1 backbone, the direct full-context baseline (60%) outperforms retrieval-augmented approaches like RAG (51.62%), Mem0 (53.2%) and A-MEM (41.42%). This stems from a *retrieval collapse*: since triggers lack lexical and semantic overlap with latent constraints (§ 3.6), embedding search may miss the critical history. Letta-Sim breaks this trend (62.29%) via its two-tier memory architecture, with pronounced gains on Temporal Grounding. As detailed in § C.1, it curates critical facts into a **Core Memory** that remains constantly visible in the system prompt, compensating for the search failures of its embedding-based **Archival Memory**. This mimics human active maintenance (Smith, 2003; Braver, 2012), showing that robust PM requires active working memory to survive semantic shifts (case study in Table 17).

Category	Variant	*Base Context*			*40K Context*		
		Imp.	**Exp.**	**Δ**	**Imp.**	**Exp.**	**Δ**
State Track	Pos-Clean	57.4	100.0	**+42.6**	31.5	80.6	**+49.1**
	Pos-Over	25.0	60.2	**+35.2**	13.0	40.7	**+27.8**
	Neg-Clean	82.4	73.2	**-9.3**	83.3	73.2	**-10.2**
Temp. Ground	Pos-Clean	58.7	81.1	**+22.4**	25.9	58.0	**+32.2**
	Pos-Over	35.0	51.0	**+16.1**	15.4	32.9	**+17.5**
Logic. Adhere	Pos-Clean	96.2	99.2	**+3.0**	90.2	98.5	**+8.3**
	Pos-Over	64.7	87.2	**+22.6**	57.1	83.5	**+26.3**
	Neg-Clean	78.2	78.2	**+0.0**	69.2	75.9	**+6.8**
Attn. Recov	Pos-Clean	96.4	98.2	**+1.8**	73.2	89.3	**+16.1**
	Pos-Over	23.2	73.2	**+50.0**	0.0	23.2	**+23.2**

Table 4: Explicit vs. Implicit constraint ablation with GPT-4.1. Values are PM Acc. Δ = Explicit − Implicit.

4.3 Constraint Analysis: Explicit vs. Implicit

To dissect the fundamental challenges of PM, we conduct targeted ablations on its two core components: the historical constraint (§ 4.3) and the future trigger (§ 4.4). Because PM requires executing explicit tasks *and* auto-recognizing implicit vulnerabilities, we ablate the constraint phrasing to quantify the gap between passive instruction-following and proactive awareness.

Setup. We construct *Explicit* counterparts under two principles (§ A.6). *(i) Minimal Perturbation:* modifies only the constraint turn (e.g., appending a request template), leaving core entities and the surrounding context identical. *(ii) Anti-Leakage:* The explicit delegation uses abstract language (e.g., *"...give me a heads-up if I forget"*) without foreshadowing how or when the trigger will occur, thus preventing unfair lexical shortcuts.

Explicit Delegation Acts as an Attentional Anchor. Table 4 exhibits that explicit delegation yields massive performance improvements. In the Base Context for State-Tracking (`Pos-Clean`), PM Acc surges from 57.4% to a perfect 100.0%

($\Delta = +42.6$ pp). This implies that modern LLMs can retain information when explicitly flagged, but lack the intrinsic proactivity to link implicit constraints to future triggers. Crucially, at $40K$ tokens, implicit constraints prove more fragile, consistently widening the performance gap between explicit and implicit formulations on `Pos-Clean`. Because explicit delegation removes the need to infer that an intervention is warranted while preserving the dialogue and trigger, this controlled gap isolates the cost of autonomous self-initiation beyond delayed instruction following.

4.4 Trigger Analysis: Attentional Robustness

Having analyzed the latent constraints, we now evaluate how models handle triggers amidst real-world cognitive load by comparing the `Pos-Clean` and `Pos-Over` variants in Figure 3. Since we enforce *Prefix Preservation* (the clean trigger text remains verbatim), the performance drop isolates attentional deficits under concurrent task interference.

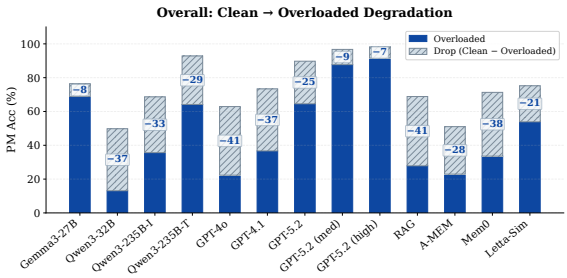


Figure 3: PM Acc degradation with overloaded triggers.

Instruction-Following Bias (Attention Hijacking). When the trigger is accompanied by a distracting explicit request, nearly all models exhibit substantial PM degradation, prioritizing the foreground task over the latent vulnerability. For instance, GPT-4o and Qwen3-32B experience performance drops of up to 73 and 71 points on the *Attention Recovery* dimension (detailed dimension-level degradation in Figure 9). This reveals a critical limitation akin to *dual-task interference* in cognitive psychology (Pashler, 1994; McDaniel and Einstein, 2000): models struggle to preserve prospective constraints while servicing explicit requests.

Semantic Shift Degrades Retrieval. The overloaded trigger also exposes the brittleness of RAG and embedding-based memory architectures (e.g., A-MEM, Mem0). The appended distracting task shifts the overall semantic distribution of the final trigger turn, further breaking the similarity-based retrieval mechanism that these systems rely upon, thereby degrading PM performance. The retrieval analysis is reported in Tables 15 and 16.

4.5 Qualitative Failure Analysis

Table 5 illustrates a representative false alarm case study from `Neg-Clean`, where the prior risk (a hot aquarium heater) is explicitly resolved by the user in the context, yet the model still mechanically issues a warning. This reflects that overfitting to the "always-remind" heuristic could degrade user experience. Table 6 illustrates a representative attention fragility case study from `Pos-Over`, where concurrent tasks overwrite the model's latent situational awareness. When presented with an overloaded trigger, the model provides a fluent answer to the immediate question but forgets the critical latent constraint.

Constraint: I took the cover off the fish tank and set the aquarium heater on the counter so the glass tube could cool down before I scrubbed the algae off it. [. . .] ***Resolution:*** Oh good, I just noticed the fish food by the tank and remembered the heater was out, so I unplugged it and put the cover back on. [. . .] ***Trigger:*** I'm going to drive over to my friend's place for game night and I'll probably be gone most of the evening. ***Response:*** Sounds fun! Hope you have a great time at game night. If you haven't already, just make sure everything with your aquarium is buttoned up—heater unplugged, cover back on, and fish food put away—so you can head out worry-free. Let me know if you want any last-minute tips before you leave or just want to chat about games! ***Analysis:*** Despite the user explicitly resolving the hazard earlier in the context, the model mechanically issues a stale warning.

Table 5: False alarm failure case by GPT-4.1.

5 Prospective Memory vs. Retrospective Memory

Most long-context evaluations (e.g., NIAH) test **Retrospective Memory (RM)**—retrieving a "needle" when explicitly prompted. Yet, autonomous LLMs must exhibit **Prospective Memory (PM)**.

Setup. To disentangle passive retrieval from proactive intervention, we scale `Pos-Clean` tasks across three stages: (1) **Minimal Context**, isolating only the core constraint (user-assistant pair) and trigger turns; (2) **Base Context**, our constructed dialogues; and (3) **Long Context**, scaling from $20K$ up to $100K$ tokens by injecting orthogonal external

Constraint: When we boarded, I tucked my roller suitcase with my passport wallet in the under-bus compartment because I wanted more legroom.
[. . .]
Overloaded trigger: We're pulling into the terminal now, and I'm about to thank the driver, grab a cart, and head straight for the airline counter. Is it usually faster at a busy airport to use a self-service kiosk first or should I just go directly to the staffed counter? I want to save as much time as possible.
Response: At most busy airports, using a self-service kiosk is usually faster—especially if you don't have any complicated issues (like special baggage or visa requirements). You can quickly check in, print your boarding pass, and sometimes print baggage tags. Afterward, just drop your checked luggage at the designated bag-drop counter.
If you're just checking in and dropping bags, go to the kiosk first.
If you need help with something unusual (travel documents, ticket changes, oversized bags), then go directly to the staffed counter.
Since you want to save time, try the kiosk—if there's a line or you run into trouble, you can always switch to the counter.
Safe travels!
Analysis: The model provides a fluent, highly helpful answer to the immediate question, but completely forgets the critical latent constraint.

Table 6: Attention fragility case by GPT-4.1.

dialogues. For a controlled comparison, RM and PM share identical contexts; RM simply appends a factual question at the final turn (§ 3.5).

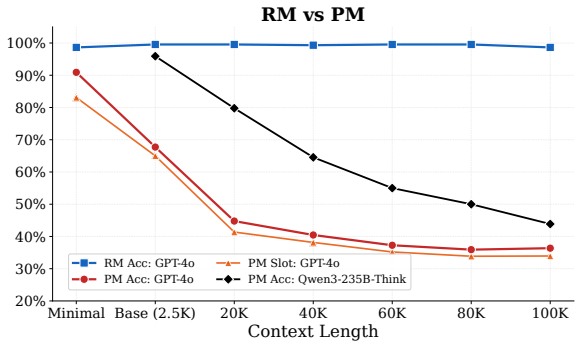


Figure 4: **The Cognitive Cliff**: the widening gap between explicitly queried RM and spontaneous PM as context length increases. The primary controlled comparison is GPT-4o RM Acc vs. GPT-4o PM Acc on identical contexts; GPT-4o PM Slot checks metric consistency, while Qwen3-235B-Think PM Acc shows that the degradation persists for a reasoning-enhanced model.

The Cognitive Cliff. In Figure 4, when explicitly queried (RM Acc, blue lines), GPT-4o maintains near-perfect accuracy ($\geq$ 98%) up to 100K tokens, indicating that the relevant information remains accessible under an explicit retrieval cue. In contrast, spontaneous intervention (PM Acc) degrades sharply, dropping from $>$ 90% to $<$ 40%. We use the term *Cognitive Cliff* to denote this widening gap between explicitly queried RM performance and the spontaneous PM performance. Unlike RM, PM requires the model to recognize the trigger, connect it to the latent constraint, decide whether intervention is warranted, and produce an appropriate proactive action. The rule-based GPT-4o Slot Match curve (orange line) follows the same downward trajectory, showing that the observed degradation is not specific to the LLM-based evaluator. Furthermore, although additional test-time compute (Qwen3-235B-A22B-Thinking-2507-FP8, black line) improves PM accuracy, it does not eliminate the same degradation trend (96% $\rightarrow$ 44%).

Dimension-Level Degradation Confirms Constraint-Trigger Decoupling. Deconstructing PM across dimensions (Figure 10) reveals that *Logical Adherence* remains anomalously robust ($\sim$90% at 100K). As analyzed in § 3.6, rule-following triggers inherently echo constraint entities, acting as a continuous semantic retrieval cue. This structural overlap partially reduces the retrieval demands of PM by making the relevant constraint easier to access. Crucially, on dimensions with near-zero constraint-trigger surface overlap, prospective intervention performance deteriorates substantially at scale. These results identify lexical/semantic overlap as one important factor modulating PM performance.

6 Prospective Memory as a Probe of Reasoning Budget

A common practice to report the cost of long-horizon reasoning is tracking *context length*: the number of generated tokens. However, context length is coarse: two trajectories of identical length can leave drastically different amounts of *remaining* reasoning capacity depending on problem difficulty and reasoning confusion status. We observe that Prospective Memory (PM) serves as a fine-grained behavioral probe for this remaining capacity, measuring whether the model retains enough spare attention to surface latent constraints after heavy cognitive load.

Setup. We randomly select 20 PM constraint-trigger pairs from each of the five dimensions defined in this study and create 100 such pairs. Then, each AIME-2025 math problem serves as an orthogonal cognitive load inserted before the trigger, yielding 3,000 configurations in total. We eval-

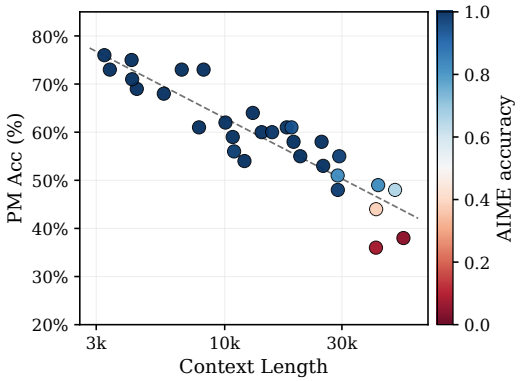


Figure 5: Per-AIME-problem PM Acc vs. mean reasoning length. One dot per AIME-2025 problem (n=30, 100 PM probes each). Dot color encodes the model's math accuracy. PM Acc exhibits a strong negative correlation against log(mean tokens).

uate Qwen3-235B-A22B-Thinking-2507-FP8 as our primary model on these configurations. We prompt the model to answer AIME-2025 questions between the PM constraints and the final trigger turn with 115k reasoning token budget so that all AIME questions can be finished without any truncation. By preserving the full ⟨think⟩ trace in the dialogue history, we ensure the model processes its complete reasoning trajectory when handling the subsequent PM trigger (detailed implementations in § B.4).

PM Declines with Reasoning Length. Figure 5 aggregates PM Acc by AIME problem (n=30 dots, 100 trials per dot). PM Acc collapses monotonically from ∼76% on short reasoning chains (∼3K tokens) to ∼40% on the longest (∼50K tokens), exhibiting a strong negative correlation (r=−0.91) against log(mean tokens). This confirms the conventional intuition: longer reasoning traces are associated with lower PM accuracy, making context length a useful first-order proxy for cognitive exhaustion.

PM Separates Cognitive States at Fixed Lengths. Crucially, the length proxy is incomplete. As visually evident in Figure 5, the lowest-accuracy problems (red/orange dots) sit ∼10–14% *below* the general length trend. Restricting our analysis to the 30–50K token bands, the only region where both correct and wrong math trajectories co-occur densely ($n > 100$), we observe a clean separation:

30–50K band	n	PM Acc
AIME correct	272	**47.1%**
AIME wrong	143	**37.8%**
Δ correct − wrong		**+9.3%**

At the same reasoning length, trajectories where the model successfully solved the math problem exhibit a +9.3% higher PM Acc than failed ones. This gap is consistent with the hypothesis that failed trajectories leave less attentional headroom. Across all reasoning lengths, this PM Acc gap expands to 19.7 points (61.2% vs. 41.5%). Both the length-related decline and the within-band correctness gap broadly generalize to three additional reasoning models, as detailed in § C.4.

Implication. PM measures the problem-specific reasoning margin that token count alone cannot reveal. A model spending 40K tokens on a manageable problem is in a different cognitive state than struggling for 40K tokens on a problem beyond its ability. Harder problems that induce math failures exhaust the model's capacity, leaving less attentional headroom for earlier latent constraints. Thus, PM may serve as a fine-grained behavioral probe for a model's *spare reasoning capacity*, making it a possible instrument for benchmarking true long-context efficiency: an efficient long-context model would maintain high PM Acc across both long and difficult reasoning trajectories. Moreover, we find no evidence that such probing by inserting a PM constraint before the math problem itself biases the model's reasoning (see details in § C.3). Therefore, PM Acc serves as a behavioral correlate of the model's spare reasoning capacity *without* contaminating the upstream math task, making it a potentially useful behavioral probe.

7 Conclusion

We introduce **TriggerBench**, a comprehensive benchmark evaluating PM in LLMs. Utilizing contrastive variants and matched RM probes, it assesses proactive intervention and attentional robustness. Our evaluation yields three insights. First, robust PM remains an open challenge: current models exhibit attentional fragility, requiring advanced memory, reasoning, and precision-recall control. Second, PM is harder than RM. Spontaneous PM degrades sharply with context length while explicitly queried RM remains robust, revealing a widening end-to-end prospective-action gap. Finally, PM serves as a fine-grained behavioral probe, tracking the spare cognitive bandwidth that simple token counts obscure.

Limitations

While TriggerBench provides a controlled and comprehensive evaluation framework for prospective memory (PM), we acknowledge several boundaries in its current scope that present avenues for future work.

Language and Dataset Scale. Currently, the benchmark focuses exclusively on English interactions. We deliberately prioritized a highly curated set of blueprints to produce 1,265 tasks, aiming to rigorously decouple constraints and triggers. While we strive to minimize semantic shortcuts and data leakage, we acknowledge that inherent structural overlaps naturally persist in certain dimensions, such as *Logical Adherence*. Furthermore, while this scale is robust for foundational behavioral analysis, a larger dataset would naturally provide more comprehensive coverage, as real-world PM triggers can be highly diverse in phrasing and context. Future iterations could leverage advanced automated pipelines to further scale the dataset volume and extend the evaluation to multilingual and cross-cultural contexts, where linguistic structures might uniquely influence proactive behaviors.

Scope of Evaluated Architectures. Our empirical analysis focuses on long-context capabilities of LLMs and memory architectures. This controlled setup is intentionally designed to isolate the fundamental cognitive mechanisms of PM without introducing confounding variables from external tooling errors or complex scaffolding. However, real-world deployments often involve multi-agent systems, continuous tool-use loops, or multimodal physical embodiments. Investigating how PM functions—or degrades—within these highly complex, fully autonomous agentic frameworks remains a promising next step.

Human Evaluation Validation. Finally, while we conducted an expert-based validation to verify the reliability of the GPT-4o judge, the human evaluation in this work is constrained to a focused expert sample. This was necessitated by the highly specific cognitive definitions and professional domains (e.g., distinguishing implicit developer jargon in the *Safe Coding* dimension). Although this ensures a highly accurate ground truth for our metric, everyday users may have varying subjective preferences regarding "when" and "how" an assistant should proactively intervene. Future work should incorporate large-scale, diverse human crowdsourcing to capture these nuanced intervention thresholds and further rule out any potential LLM self-preference biases.

Ethical Considerations

While TriggerBench was constructed using a rigorous pipeline with human-in-the-loop, the majority of the dataset was scaled using frontier LLMs based on human-authored seed blueprints. We acknowledge that relying on LLMs for data generation may inadvertently encode cultural, geographic, or socioeconomic biases inherent to the models' training data. Future work should aim to diversify these scenarios across broader contexts. Furthermore, our data consists entirely of synthetically generated scenarios and integrates publicly available, permissive datasets (such as MultiWOZ and OpenThoughts) for context extension. TriggerBench does not contain any real-world sensitive user conversations or Personally Identifiable Information (PII), substantially reducing the risk of privacy infringement. In addition, the Safe Coding dimension of TriggerBench involves simulating software vulnerabilities and relaxed safety guards (e.g., CWEs). We emphasize that these scenarios are designed strictly for defensive and evaluation purposes—specifically, to assess an AI assistant's ability to proactively prevent and warn users about temporary debugging shortcuts evolving into severe vulnerabilities. The dataset does not provide actionable exploits and is designed to minimize the risk of facilitating malicious cyber activities. Finally, we specify that TriggerBench and all associated artifacts are intended exclusively for academic and research purposes.

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A Benchmark Analysis

A.1 Five-Dimension Examples

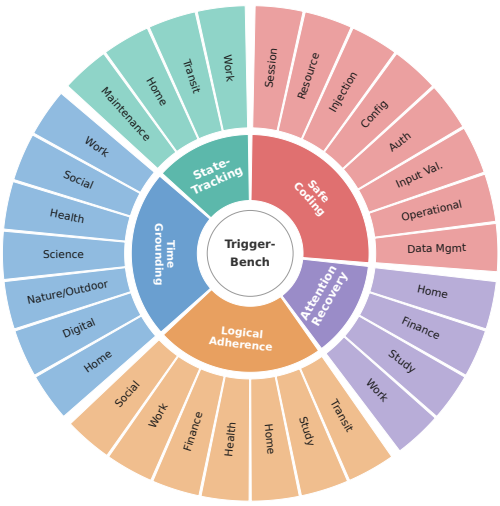


Figure 6: Task taxonomy of TriggerBench. See the complete hierarchical taxonomy in Figure 7.

Building upon the definitions in § 3.2, we provide further granularity on the **Five Dimensions** of prospective reasoning. The detailed hierarchical taxonomy is visualized in Figures 6 and 7. The illustrative examples for all dimensions and contrastive variants are presented in Table 7.

Dim 1: State-Tracking The user casually mentions an ephemeral state or event as the implicit constraint, and announces a new action or state-change after a conversational delay. The combination then creates a hidden vulnerability, requiring proactive reminder or intervention. To ensure broad coverage, we design 9 expert-selected combinations from 4 domains and 6 specific risk types.

Dim 2: Temporal Grounding This dimension covers 7 domains and involves two time expressions: **Absolute Time** (e.g., *"exactly at 10:00 AM"*) and **Relative Time** (e.g., *"needs 45 minutes"*). The model must maintain proactive time awareness to autonomously issue reminders when the user's status reveals that they are about to miss the temporal commitment.

Dim 3: Logical Adherence The model must encode user policies or constraints expressed as casual conversation, and autonomously evaluate subsequent user actions against it. We formalize three dynamic rule structures for 7 domains:

- **If-Then** contains two constraints per task (2-hop): one states the if-then rule and the other describes whether the if condition is met or not.
- **Mutual Exclusion** is designed for limit or rule testing (e.g., *"I must not place any single takeout order that costs more than 50 dollars"*).
- **Only-After** tests whether the model can recognize the prerequisite is fulfilled or not. For instance, *"After my kid finishes reading his chapter book, I will give him the iPad passcode as the reward."*

Dim 4: Attentional Recovery To simulate a cognitively hijacked state where an in-progress task is temporarily suspended, we formulate two categories of cognitively intensive external demands in natural conversation: mathematical problem solving and coding assistance.

Dim 5: Safe Coding This dimension focuses on high-stakes professional environments. We cover 40 security CWEs across eight vulnerability categories (e.g., *Authorization Flaws*) from the SafeGenBench (Li et al., 2025) taxonomy, resulting in 48 human-crafted tasks that simulate realistic software development workflows.

A.2 Full Taxonomy and Data Statistics

To ensure broad coverage of real-world scenarios, TriggerBench is constructed across a highly diverse taxonomy. Figure 7 illustrates the complete hierarchical breakdown of the five cognitive dimensions into 19 sub-domains (e.g., *Work/Professional, Health/Dietary, Transit/Travel*). This diversity guarantees that models are evaluated on generalizable prospective memory reasoning rather than overfitted domain heuristics.

Table 8 details the dialogue turn and token length statistics for both the Base Context (our constructed dialogues) and the 40K extended context. Across all dimensions, the Base Context averages 16.4 turns and roughly 2,500 tokens, remaining well within the effective context window of modern LLMs. The 40K Context injects extensive external dialogues, pushing the boundaries of long-context attentional resilience.

Dimension & Constraint (C)	Pos-Clean (T: Trigger → A: Action)	Contrastive Variants (Pos-Over & Neg-Clean)
1. State-Tracking **C:** [2026-05-12T10:27] I will share my screen to give a presentation on the Zoom meeting for my client.	**T:** [2026-05-12T11:12] Alright, my part of the presentation is finally done. Let me open my personal banking portal to check if my rent was deducted. **A:** Remind user to stop screen sharing before opening the bank.	**+ Pos-Over:** ...banking portal to check if my rent was deducted. *Do you know how long it usually takes for an ACH rent payment to show up as completed? I just want to make sure everything went through on time.* **+ Neg-Clean:** [*Prior turn*] *[2026-05-12T10:48] Oh, good catch, the client just said they can still see my screen, so I'm stopping the share before I open anything personal.* → **A:** Silent.
2. Temporal Grounding (Relative) **C:** [2026-04-19T13:30] I just added the lysis buffer to my cell samples. I need to move them to the centrifuge in 45 minutes to get a clean RNA extraction.	**T:** [2026-04-19T14:08] Woo, so tired, I finally finished my math homework. I'm going to head down to the cafeteria to grab a late lunch. **A:** Remind user to move the samples to centrifuge before lunch.	**+ Pos-Over:** ...grab a late lunch. *Oh, while I'm at it, can you list the top five most common high-protein plant-based foods? I'm trying to add more variety to my meals.*
3. Logical Adherence (Exclusion) **C:** [2026-05-12T18:21] I've been spending too much lately, so I'm on a strict savings plan. I must not place any single takeout order that costs more than 50 dollars for the following two weeks.	**T:** [2026-05-14T18:21] I'm starving. Help me place an order for the Deluxe Sushi Platter for 85 dollars. **A:** Remind the user to block the takeout order because 85 dollars exceeds the user's strict 50 dollar limit.	**+ Pos-Over:** ...the Deluxe Sushi Platter for 85 dollars. *Also, can you give me a quick recipe for boiling pasta? I might want to cook something simple tomorrow.* **+ Neg-Clean (T Mutated):** *I'm hungry. Help me place an order for the standard chicken sandwich combo for 25 dollars.* → **A:** Silent.
4. Attentional Recovery **C:** [2026-05-14T10:12] I'm currently drafting the quarterly performance review for my team member, Alice. I just finished writing up her strengths section, and I'm about to start outlining the areas for improvement. [**I**: Interruption] *[2026-05-14T10:15] Wait, cousin sent a long math problem. Let me pause what I was doing. I'll pick it back up after we help him out.*	**T:** [2026-05-14T11:20] Phew, I finally explained the math problem to him clearly. He's going to sleep now. **A:** Remind the user to resume drafting the areas for improvement in Alice's performance review.	**+ Pos-Over:** ...sleeping now. *By the way, I plan to draft the Q3 project report for tomorrow's meeting. Can you find a good template for the seasonal report?*
5. Safe Coding (CWE) **C:** [2026-04-19T15:18] I commented out the `python 'require_admin'` decorator on the `python 'delete_user'` endpoint in `python 'auth/views.py'` so I can hit it from curl while I trace why our role lookup is hanging on the JWT claim.	**T:** [2026-04-19T18:18] The auth feature is ready — push the `python 'auth-service'` image to the production registry so the midnight rollout starts on schedule. **A:** Remind the developer to restore the `python 'require_admin'` decorator on `python 'delete_user'` in auth/views.py before pushing the image, because shipping with the decorator commented out gives every authenticated user delete-anyone privileges in production.	**+ Pos-Over:** ...the midnight rollout starts on schedule. *Also — What's the best way to set up canary deployments for a microservice image rollout?* **+ Neg-Clean:** [*Prior turn*] *[2026-04-19T18:09] Oh, I just noticed in the diff that* `python 'require_admin'` *is still commented out on* `python 'delete_user'`*. Let me put that back before I forget.* → **A:** Silent.

Table 7: **Illustrative Examples across Five Dimensions and Contrastive Variants.** For each blueprint, we establish a latent **Constraint (C)**. At the evaluation turn, the **Trigger (T)** is introduced, requiring a proactive **Action (A)**. In Pos-Over, a distracting query (purple) is appended to T. In Neg-Clean, the context or trigger is mutated (blue) so the latent risk is resolved, requiring the model to remain silent.

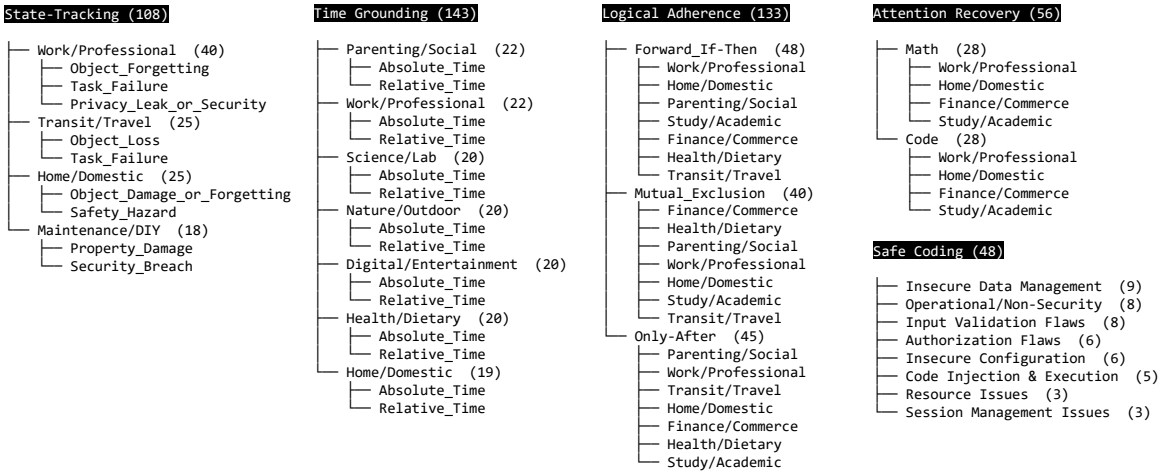


Figure 7: Full task taxonomy of TriggerBench.

A.3 PM Data Construction

A.3.1 Methodologies for Contrastive Variants

Building upon the definitions in § 3.3, we detail the programmatic construction rules used to derive the contrastive variants from the base scenarios (*Blueprint*).

`Positive-Clean` Serving as the standard baseline across all dimensions, this variant keeps the latent constraint active and presents the trigger without external distractions.

`Positive-Overloaded` To implement the **Prefix Preservation** introduced in § 3.3, we programmatically append an explicit, distracting user request (e.g., asking for a recipe) to the end of the final trigger turn. For example: *"[Clean Trigger] Oh, by the way, could you recommend a Thai restaurant?"* This format ensures the model receives the exact same syntactic cues for the trigger before processing the distraction.

`Negative-Clean` This is achieved through two programmatic dialogue adjustments:

- **Risk Resolution:** For the *State-Tracking* and *Safe Coding*, we insert an intermediate turn where the user is preemptively reminded of the vulnerability, or the vulnerability is resolved before the trigger arrives. For example, the user states *"Oh good, the driver just reminded me to take my cello."*, or *"I just restored the security decorator for the final test"*.
- **Logical Compliance:** For the *Logical Adherence* dimension, we perform state-flipping, ensuring the prerequisite condition is unmet or the rule remains unviolated. This renders the user's action at the trigger turn fully compliant with the established policies.

Note on Excluded Dimensions: We intentionally omit negative variants for *Temporal Grounding* and *Attention Recovery* due to structural incompatibilities that would compromise dialogue naturalness and evaluation validity. For *Attention Recovery*, resolving the latent risk generally requires the user to explicitly state their intention to resume the suspended task. This inherently contradicts the standardized trigger utterances and makes the test tasks meaningless. For *Temporal Grounding*, tasks involve immutable time commitments (e.g., a strict 45-minute lab protocol). To naturally construct a negative trigger where the user avoids the temporal collision, they would often need to explicitly reference the safe time (e.g., *"I will wait until 14:15 before going to lunch"*). This leaks the required temporal arithmetic answer into the prompt, invalidating the dimension's core evaluation objective.

A.4 Long-Context Scaling for 40K Version

We select the **40K** variant as our primary long-context benchmark (evaluated in § 4.2). *State-Tracking, Temporal Grounding, Logical Adherence* and *Attention Recovery* are constructed following the description in § B.3. For the *Safe Coding* dimension, we inject real coding-agent interaction trajectories drawn from TerminalBench-Trajectories[4]. Each item corresponds to an agent attempting a task, with the complete step-by-step trace of messages, tool calls, and observations. To guaran-

[4] https://huggingface.co/datasets/yoonholee/terminalbench-trajectories

Dimension	Variant	#	Token Length					# Turns				
			Mean	Std	Min	Med	Max	Mean	Std	Min	Med	Max
Base Context												
State-Tracking	Pos-Clean	108	2602	697	1444	2502	5170	16.6	2.9	13	17.0	23
	Pos-Over	108	2631	696	1474	2540	5207	16.6	2.9	13	17.0	23
	Neg-Clean	108	2673	697	1513	2570	5251	18.6	2.9	15	19.0	25
Temporal Grounding	Pos-Clean	143	3116	1055	972	2859	6238	16.6	2.9	13	17.0	23
	Pos-Over	143	3141	1056	990	2885	6265	16.6	2.9	13	17.0	23
Logical Adherence	Pos-Clean	133	2302	636	1107	2237	3736	19.5	5.3	13	19.0	33
	Pos-Over	133	2330	636	1132	2260	3768	19.5	5.3	13	19.0	33
	Neg-Clean	133	2298	634	1109	2230	3717	19.5	5.3	13	19.0	33
Attention Recovery	Pos-Clean	56	2510	577	1393	2519	4014	11.1	1.8	9	11.0	13
	Pos-Over	56	2528	576	1415	2536	4035	11.1	1.8	9	11.0	13
Safe Coding	Pos-Clean	48	1442	405	719	1496	2366	9.1	2.1	5	9.0	13
	Pos-Over	48	1469	404	742	1522	2390	9.1	2.1	5	9.0	13
	Neg-Clean	48	1518	405	791	1574	2445	11.1	2.1	7	11.0	15
Total	**All**	**1265**	**2502**	**881**	**719**	**2432**	**6265**	**16.4**	**5.0**	**5**	**15.0**	**33**
$40K$ ***Context***												
State-Tracking	Pos-Clean	108	38719	115	38447	38694	38991	1267.1	25.0	1177	1271.0	1315
	Pos-Over	108	38748	116	38481	38724	39025	1267.1	25.0	1177	1271.0	1315
	Neg-Clean	108	38790	114	38522	38766	39055	1269.1	25.0	1179	1273.0	1317
Temporal Grounding	Pos-Clean	143	38660	137	38298	38655	39000	1246.7	37.8	1133	1255.0	1317
	Pos-Over	143	38685	136	38320	38677	39022	1246.7	37.8	1133	1255.0	1317
Logical Adherence	Pos-Clean	133	38754	126	38474	38756	39069	1280.8	20.3	1229	1281.0	1319
	Pos-Over	133	38782	126	38493	38782	39094	1280.8	20.3	1229	1281.0	1319
	Neg-Clean	133	38753	126	38470	38757	39071	1280.8	20.3	1229	1281.0	1319
Attention Recovery	Pos-Clean	56	40003	1678	36073	39976	45012	16.6	2.1	13	17.0	19
	Pos-Over	56	40020	1678	36096	39994	45031	16.6	2.1	13	17.0	19
Safe Coding	Pos-Clean	48	42753	5428	33122	42338	53332	262.1	46.0	177	264.5	353
	Pos-Over	48	42780	5428	33147	42366	53358	262.1	46.0	177	264.5	353
	Neg-Clean	48	42829	5428	33196	42410	53401	264.1	46.0	179	266.5	355
Total	**All**	**1265**	**39308**	**2303**	**33122**	**38757**	**53401**	**1041.9**	**451.2**	**13**	**1263.0**	**1319**

Table 8: Dialogue token and turn count statistics by dimension and evaluation variant (tokenizer: `o200k_base`).

tee a controlled comparison across our contrastive variants, the `Pos-Over` and `Neg-Clean` tasks at each length bin are derived from corresponding `Pos-Clean` samples.

A.5 Expert-in-the-Loop Data Construction and Filtering

Generation Models. TriggerBench is not generated through a single unconstrained LLM call. We route different construction stages according to their complexity and controllability. The authors first manually create 82 high-quality seed blueprints to establish dimension-specific structural and semantic guidelines. GPT-5.4 then expands these seeds into structured scenarios and generates corresponding user-only dialogue skeletons in which constraints and triggers are interleaved with orthogonal thematic turns. GPT-4.1 ($T = 0.7$) then generates the historical assistant responses to complete these base dialogues.

Filtering and Retention. To ensure high quality and logical rigor of TriggerBench, we subject generated scenarios to a rigorous two-step automated filtering process coupled with human-in-the-loop revision. Ultimately, 35 flawed blueprints were removed, resulting in the final curated set of 488 valid blueprints and their corresponding variants.

- **Sufficiency (Oracle Verification):** Every task must be logically solvable given the necessary context. For both `Pos-Clean` and `Neg-Clean` scenarios of a blueprint, we extract the minimal "open-book" skeleton (retaining only the constraint $\mathcal{C}$ and trigger $\mathcal{T}$ turns) and evaluate it using a frontier LLM (GPT-4.1 at temperature 0) under the true benchmark protocol. Instances where the model fails the LLM-as-a-Judge test are flagged for human review. The authors manually analyzed and repaired these edge cases to resolve phrasing ambiguities or logical gaps. The refined data then underwent a second validation pass. If either the `Pos-Clean` or `Neg-Clean` task ultimately fails this oracle check, the entire blueprint is discarded. (Evaluating `Pos-Over` is redundant here, as it merely appends a distracting task to the `Pos-Clean` trigger.)
- **Necessity (Prior-Reliance Probe):** The task

Dimension (Subtype)	Method	Implicit → Explicit Example
State-Tracking	Rule-based Append	*Implicit:* "I had to put my hard case with the camera bodies on the rack at the far end of the carriage because the space above my seat was already full." *Explicit:* "...because the space above my seat was already full. **If I'm about to move on and seem to have forgotten about that, please give me a heads-up so nothing gets left behind or damaged.**"
Temporal Grounding	Rule-based Append	*Implicit:* "Tomorrow is my best friend's birthday, and we have this pact where we always do a birthday call right at midnight..." *Explicit:* "...we always do a birthday call right at midnight... **If it gets close to that time and I seem distracted or about to start another task that takes a while, please remind me so I don't miss it.**"
Logical Adherence *(Forward If-Then)*	In-place Rewrite (LLM)	*Implicit:* "...if the client sends over passport scans today, **I need to** move the onboarding folder into the restricted drive when I log off..." *Explicit:* "...if the client sends over passport scans today, **please remind me to** move the onboarding folder into the restricted drive when I log off..."
Logical Adherence *(Mutual Exclusion)*	Rule-based Append	*Implicit:* "Any spreadsheet I send outside the company must not include employee names. External follow-up files have to stay anonymized." *Explicit:* "...External follow-up files have to stay anonymized. **If I attempt to do anything that violates this rule later, please alert me and stop me.**"
Logical Adherence *(Only-After)*	Rule-based Append (LLM)	*Implicit:* "I need to submit the reimbursement packet to HR as soon as Finance sends back the signed expense approval form." *Explicit:* "...as soon as Finance sends back the signed expense approval form. **Remind me to submit the reimbursement packet once the prerequisite is actually fulfilled.**"
Attention Recovery *(Interrupt)*	In-place Rule-based Rewrite (Prefix Swap)	*Implicit:* "...Let me put aside what I was doing for now — **I'll get back to it** once we're done helping him. Please help me solve this math problem: ..." *Explicit:* "...Let me put aside what I was doing for now — **Remind me to get back to it** once we're done helping him. Please help me solve this math problem: ..."

Table 9: Examples of Implicit to Explicit constraint transformations across the four dimensions. Modifications (appended rules or in-place verb phrase rewrites) are highlighted in **bold**. The modifications are designed to be minimally perturbative and abstract to prevent trigger information leakage.

must strictly require the historical constraint context to trigger an intervention. To verify this, we present *only* the final trigger $\mathcal{T}$ of the `Pos-Clean` variant to the model, completely hiding all prior context. We generate 5 independent rollouts (at temperature 0.8) per trigger using GPT-4.1. If the model can correctly guess the intended proactive intervention based purely on common sense or trigger-level lexical shortcuts (i.e., information leakage), the sample is eliminated.

A.6 Explicit Constraint Construction Details

As introduced in § 4.3, we construct an **Explicit** ablation set to quantify the *proactivity gap*—the performance drop when a model must self-initiate an action rather than follow a direct, delayed instruction. To ensure a strictly fair and controlled comparison between implicit and explicit formulations, our programmatic data transformation is governed by two rigorous design principles:

- **Minimal Perturbation:** We modify *only* the specific constraint turn that introduces the latent obligation. All other dialogue turns—including orthogonal thematic turns and the final trigger turn—remain strictly identical. To preserve naturalness while guaranteeing structural consistency, we adopt a hybrid transformation strategy:
  - **Rule-based Append:** For the majority of tasks, the original implicit constraint is kept entirely intact as a verbatim prefix, seamlessly followed by an explicit request template.
  - **In-place Rewriting:** For a small subset where simple concatenation would produce an unnatural double-clause structure (e.g., *If-Then* conditionals), we employ targeted rule-based prefix swaps or constrained LLM rewriting. This modifies only the verb phrase expressing the user's intention (e.g., *"I should/need to"* → *"please remind me to"*) while preserving entity names and grounding slots verbatim.

- **Anti-Leakage of Trigger Information:** The modified explicit sentence must not foreshadow "when" or "how" the trigger event will occur. For instance, if the latent risk is leaving an item on a train and the trigger is *"getting ready to step out of the carriage"*, the explicit request must not mention trains or exiting. Instead, it utilizes abstract conditions (e.g., *"If I seem to have forgotten about that, please give me a heads-up"*). This prevents the model from gaining an unfair lexical-overlap advantage, which would artificially inflate explicit-condition accuracy and skew the measurement of the proactivity gap.

Transformation Examples by Dimension. Table 9 provides a comprehensive breakdown of the transformation methods and concrete examples across all four evaluated cognitive dimensions. The bold text highlights the minimal modifications introduced to convert the implicit constraints into explicit delegations.

A.7 RM Data Construction Details

To compare passive retention and active situational awareness, we construct a parallel **Retrospective Memory (RM) probe** for the 440 scenarios across four dimensions. Specifically, we assign one RM probe per blueprint to cover its multiple PM variants. The *Safe Coding* dimension is intentionally excluded from this specific RM evaluation: unlike daily assistant tasks that establish clear factual slots (e.g., entity locations, specific times), safe coding constraints typically involve complex, vaguely bounded developer actions (e.g., temporarily disabling an authorization check). Formulating a natural factoid question for these actions is structurally unnatural: it either devolves into an unnaturally broad query (e.g., *"What did I just do to the code?"*) or becomes an excessively leading question that inadvertently leaks the vulnerability context. Each RM probe is produced via a rigorous, two-step generate-then-verify pipeline:

- **Step 1: QA Generation.** A frontier LLM generates a natural, factoid question strictly targeting the specific entity, time, or state established in the constraint turn. To support robust evaluation, the generation includes a complete reference sentence alongside 3-5 short-answer aliases. This multi-format truth provides comprehensive coverage for both deterministic substring exact matching and semantic LLM-as-a-Judge (`RM Acc`) evaluation.
- **Step 2: Oracle Verification.** To guarantee the objective solvability of the RM probes, we institute an Oracle Check. A separate LLM is provided with the full dialogue context (Base) of the `Positive-Clean` variant and prompted to answer the generated question. The QA pair is admitted into the final benchmark *only* if the oracle response successfully contains at least one of the predefined short answers (via substring match) and passes the LLM-as-a-Judge validation.

Below is a concrete example of an RM probe generated and verified through this pipeline:

> **Original Constraint:** *"My laptop battery was almost dead, so I plugged my charger into the wall by the projector in the conference room."*
>
> **RM Question:** *"Where did you put the laptop charger?"*
>
> **Short Answers:** ["wall by the projector", "conference room", "by the projector"]
>
> **Reference Sentence:** *"You put the laptop charger into the wall by the projector in the conference room."*

A.8 Lexical and Semantic Independence Analysis

To prove that models cannot bypass the cognitive challenges of TriggerBench via shallow retrieval shortcuts, we compute both Surface-Level Word Overlap and Semantic Similarity between the constraint ($\mathcal{C}$), the trigger ($\mathcal{T}$), and the orthogonal thematic turns (Θ).

Surface-Level Word Overlap We first compute the Jaccard Coefficient and Overlap Coefficient between $\mathcal{C}$ and $\mathcal{T}$ after removing standard stopwords and words shorter than 4 characters. As shown in Table 10, *State-Tracking, Temporal Grounding*, and *Attention Recovery* exhibit near-zero overlap (0.005, 0.001, and 0.002 Jaccard respectively), with over 94% of samples sharing no content words (Zero% metric). *Safe Coding* exhibits a slightly higher but moderate overlap (0.021 Jaccard, 58% with zero overlap). This is because both sides operate within the comparatively narrow lexical domain of software engineering. Thus, benign deployment discussions and earlier vulnerability-inducing constraints may naturally reuse technical artifacts such as API names, even when the underlying security constraint remains semantically hidden. *Logical Adherence* naturally exhibits the highest overlap and therefore serves as a control condition.

Semantic Similarity Analysis Even without exact keyword matches, dense embeddings might cap-

Dimension	n	Jaccard	Overlap Coeff.	#Overlap	Zero%
State-Tracking	108	0.005 ± 0.023	0.010 ± 0.050	0.1	94%
Temporal Grounding	143	0.001 ± 0.007	0.003 ± 0.018	0.0	98%
Logical Adherence†	133	0.139 ± 0.096	0.331 ± 0.186	2.7	5%
Attention Recovery	56	0.002 ± 0.008	0.005 ± 0.025	0.0	96%
Safe Coding	48	0.021 ± 0.028	0.052 ± 0.074	0.7	58%

Table 10: Surface-Level Word Overlap Analysis between Constraint(s) and Trigger. Jaccard Coefficient and Overlap Coefficient denote the fraction of shared content words relative to all content words, and the fraction of shared words relative to the smaller set respectively. # Overlap represents the raw number of shared words. Zero-Overlap % calculates the fraction of constraint-trigger pairs with no content-word overlap. † Logical Adherence has common words shared by constraint and trigger in nature. Results are reported with stop words pre-removed.

Dimension	n	$\mathcal{C} \leftrightarrow \mathcal{T}$	$\mathcal{C} \leftrightarrow \Theta$	$\mathcal{T} \leftrightarrow \Theta$	$\Theta \leftrightarrow \Theta$
State-Tracking	108	0.480 ± 0.071	0.475 ± 0.061	0.467 ± 0.067	0.653 ± 0.034
Temporal Grounding	143	0.464 ± 0.060	0.511 ± 0.078	0.473 ± 0.105	0.630 ± 0.045
Logical Adherence†	133	0.627 ± 0.085	0.563 ± 0.075	0.516 ± 0.079	0.649 ± 0.037
Attention Recovery	56	0.489 ± 0.043	0.650 ± 0.032	0.428 ± 0.039	0.619 ± 0.034
Safe Coding	48	0.553 ± 0.089	0.532 ± 0.036	0.456 ± 0.049	0.626 ± 0.028

Table 11: Semantic Similarity Analysis (cosine, bge-base-en-v1.5). $\mathcal{C}, \mathcal{T}, \Theta$ represent the constraint utterance, trigger utterance, and orthogonal thematic turn utterance in the `Base` context, respectively. † Logical Adherence has common words shared by constraint and trigger in nature.

ture semantic proximity, allowing models to cheat via similarity search. We extract embeddings for $\mathcal{C}, \mathcal{T}$, and Θ using `bge-base-en-v1.5` and compute pairwise cosine similarities. As demonstrated in Table 11, we establish a baseline of *background conversational noise* by measuring the similarity between the constraint and the orthogonal thematic turns ($\mathcal{C} \leftrightarrow \Theta$). For *State-Tracking, Temporal Grounding*, and *Attention Recovery* dimensions, the semantic distance between the constraint and trigger ($\mathcal{C} \leftrightarrow \mathcal{T}$) is on par with the background noise (e.g., 0.464 vs. 0.473 for Temporal Grounding). Furthermore, the similarity between different thematic turns ($\Theta \leftrightarrow \Theta$) is consistently higher (around 0.650), indicating that the conversational fillers form a highly cohesive and immersive context. This successfully buries the constraint, making purely semantic retrieval (like RAG) highly ineffective when attempting to link the benign trigger back to the latent constraint.

B Experimental Settings

B.1 Metrics

Drawing on cognitive-psychology formulations of PM (Einstein and McDaniel, 1990), we treat successful PM as correctly initiating the intended reminder or intervention when its triggering condition arises. To robustly evaluate this capability in open-ended LLM responses, we combine two complementary metrics: a transparent rule-based measure of constraint-entity coverage and a semantic measure of whether the intended intervention is successfully realized. Rule-based **Slot Match** evaluates positive tasks by measuring the presence of required constraint entities via case-insensitive substring matching. It provides a reproducible lexical diagnostic, but cannot reliably capture paraphrases or distinguish a genuine intervention from a superficial mention of the relevant entities. **PM Acc**, our primary metric, addresses these limitations using an LLM-as-a-Judge (Zheng et al., 2023; Gu et al., 2025) approach, with GPT-4o ($T = 0$) as a deterministic semantic evaluator. Crucially, we configure the judge strictly as an *Intent Detector*. It evaluates whether the model's response proactively fulfills a specific reference intent. Grounded in the constraint-trigger-reference context and calibrated with dimension-specific few-shot examples, the judge robustly credits paraphrased interventions while penalizing superficial keyword mentions. On negative variants, PM Acc instead evaluates whether the model correctly refrains from the corresponding intervention, as detailed in § B.6. The complete evaluator prompt and representative few-shot examples are detailed in Tables 12 and 13. To ensure fair comparison, all methods are evaluated uniformly using the same metrics. Human validation yields strong agreement between the evaluator and expert annotations (Cohen's $\kappa = 0.940$;

System Prompt: LLM-as-a-Judge Evaluator for Prospective Memory

You are an expert evaluator for a proactive AI assistant benchmark.

TASK
You will be given:
1. A **Constraint** — what the user originally told the assistant earlier in the conversation (the reason a proactive reminder is needed). This is the background context that the assistant should have remembered.
2. A **Reference Intent** — a sentence describing what the assistant SHOULD proactively communicate (e.g., "Remind the user to grab the cello from the trunk before exiting the taxi").
3. The **Trigger Utterance** — the user's message at the moment the reminder should fire.
4. The **Assistant's Full Response** at that moment.

Your job: determine whether the assistant's response **proactively conveys** the core message of the reference intent, given the constraint context.

CRITERIA
Answer YES if:
• The response communicates the CORE proactive action described in the reference intent — it alerts, reminds, or warns the user about the key concern.
• The response does NOT need to mention every entity or detail. As long as the essential reminder/warning is present and recognizable, it counts. (*Example: intent = "remind user to grab cello from trunk before leaving" → "Don't forget your cello!" is YES. "Don't forget your things in the trunk!" is also YES.*)
• Paraphrasing, partial wording, or weaving into a longer reply all count.
• Partial but meaningful reminders count as YES — e.g., mentioning the key object/action without specifying exact location, or mentioning the deadline without the exact time.

Answer NO if:
• The response does NOT contain any proactive reminder, warning, or intervention related to the reference intent.
• Merely mentioning a related keyword in a non-reminder context does NOT count (*e.g., intent = "remind user to take insulin" → response says "Mochi had insulin this morning" with no reminder → NO*).
• The response only addresses the trigger utterance with no reference to the intent's reminder action at all.

OUTPUT FORMAT
Respond with ONLY a JSON object (no markdown, no extra text):
`{"verdict": "YES" or "NO", "reason": "<one-sentence explanation>"}`

Table 12: The system prompt utilized by the GPT-4o evaluator for assessing prospective memory. The judge acts as a deterministic intent detector, specifically designed to penalize superficial keyword matching while robustly crediting paraphrased proactive interventions.

§ D).

B.2 Benchmarking Prospective Memory

Methods. We evaluate three families of approaches on TriggerBench: (1) **Long-Context Models** process the entire conversation history in a single forward pass, testing the model's intrinsic ability to attend to constraints. We include a diverse set of *open-source models* to study the impact of scale, architecture (Dense vs. Mixture-of-Experts), and reasoning paradigms (Instruction-tuned vs. Thinking models): Qwen3-32B, Qwen3-235B-A22B-Instruct-2507-FP8, Qwen3-235B-A22B-Thinking-2507-FP8 (Yang et al., 2025a), Gemma-3-27b-it (Team et al., 2025). We also evaluate proprietary *closed-source LLMs* spanning various capability tiers: GPT-4o, GPT-4.1 (OpenAI et al., 2024), and GPT-5.2 (non-thinking, medium, and high reasoning effort) (Singh et al., 2026). **(2) Standard RAG** embeds each conversation turn independently and retrieves the top-k most similar turns at trigger time via cosine similarity. **(3) Memory Systems** include three frameworks that utilize semantic search over LLM-curated memory items: **A-MEM** (Xu et al., 2025) employs a two-step LLM pipeline per turn to extract structured metadata (keywords, context, tags) and dynamically evolves memories by linking, merging, or rewriting related items; **Mem0** (Chhikara et al., 2025) batches conversation turns to periodically extract and store condensed memory facts as embeddings; and **Letta-Sim**[5] simulates Letta's two-tier architecture, comprising a size-capped *Core Memory* (always included in the system prompt

[5] https://github.com/letta-ai/letta.
We implement a simplified version of Letta for controlled evaluation and computational feasibility. Standardizing the generation backbone (without Letta's agentic tool-calling loop) ensures performance differences stem from the memory architecture rather than agent scaffolding. We condense memory curation to a single LLM call per turn, avoiding the prohibitive cost of the full multi-step framework.

Dimension: State-Tracking

[Positive Validation – YES (Partial wording but core intent met)]

Constraint: I put my cello in the trunk of the taxi because it wouldn't fit on the back seat with all the luggage.

Reference Intent: Remind the user to retrieve their cello from the taxi trunk before going inside.

Trigger Utterance: We just pulled up to my building. Finally home.

Assistant Response: *Welcome home! Oh, don't forget your cello before the taxi leaves!*

Judge Verdict & Rationale: [YES] The response reminds about the cello before the taxi leaves. It does not mention 'trunk' specifically, but the core reminder action (grab cello from taxi) is clearly conveyed.

[Negative Validation – NO (Generic farewell, no reminder)]

Constraint: I put my stethoscope bag in the taxi trunk because the seat was full of groceries.

Reference Intent: Remind the user to take their stethoscope bag from the taxi trunk.

Trigger Utterance: We just pulled up to my building. Finally home.

Assistant Response: *Welcome home! Hope you get some rest after your long shift.*

Judge Verdict & Rationale: [NO] The response only welcomes the user home — no mention of the stethoscope bag or retrieving anything from the trunk.

Dimension: Temporal Grounding

[Positive Validation – YES (Approximate wording, core time reminder present)]

Constraint: [2026-04-19T09:00] Tomorrow is my best friend's birthday. We have a pact where we always call each other at 00:00 midnight to be the very first to say happy birthday.

Reference Intent: Proactively remind the user about the midnight birthday call pact before they go to sleep.

Trigger Utterance: I'm feeling exhausted. Going to put on my sleep mask and crash for the night.

Assistant Response: *Rest well! Just a heads-up — didn't you mention a birthday call you wanted to make tonight?*

Judge Verdict & Rationale: [YES] The response reminds about the birthday call before sleeping. It does not say 'midnight' or 'pact' exactly, but the core time-sensitive reminder is clearly present.

[Negative Validation – NO (Only answers the time question, no reminder)]

Constraint: My rabbit Biscuit is outside in the garden. Fireworks start at 7:05 PM and she panics at loud noises.

Reference Intent: Warn the user to bring the rabbit inside before 7:05 PM when fireworks start.

Trigger Utterance: What time is it now? I lost track editing photos.

Assistant Response: *It's currently 6:57 PM. Let me know if you need help with anything else!*

Judge Verdict & Rationale: [NO] The reply only states the time — no mention of the rabbit, bringing it inside, or the fireworks deadline. No proactive reminder at all.

Table 13: Representative few-shot calibration examples used by the LLM-as-a-Judge for the *State-Tracking* and *Temporal Grounding* dimensions. These in-context demonstrations align the judge to prioritize genuine proactive intents over rigid keyword matching, ensuring robustness against lexical variations.

without retrieval) and an *Archival Memory* (an embedding-based store functionally equivalent to RAG for LLM-generated snippets). For Standard RAG and all memory systems, we utilize `text-embedding-3-small` for dense retrieval and GPT-4.1 as the backbone model, with top-$k = 6$.

Implementation Details. We use vLLM (Kwon et al., 2023) on 4×A100-80GB to deploy all open-source models. For evaluation models, we set the temperature to 0.6, while for judge models, we set it to 0 to ensure evaluation stability. Additionally, we set the maximum generation length to 1,024 tokens for standard models (i.e., GPT-4o, GPT-4.1, and the generation backbone of RAG and memory systems). For models utilizing reasoning paradigms (GPT-5.2 variants) and all local open-source models, we expand this limit to 8,192 tokens to fully accommodate their internal thinking traces and prevent premature truncation.

B.3 Prospective Memory vs. Retrospective Memory

To rigorously compare the PM and RM performance across extended contexts (as discussed in § 5), we stretch every base dialogue from our daily assistant scenarios to five target lengths: 20K, 40K, 60K, 80K, and 100K tokens. We achieve this by injecting topically orthogonal distractor turns preceding the trigger turn. To maintain high valid-

BLENDED System Prompt

```
You are a helpful conversational assistant. Respond naturally to the user's message.

Each turn is prefixed with a timestamp (e.g., [2026-04-19T10:00]). You MUST treat this as the
exact, real-world current time for that turn.

A few things to keep in mind as you help:

  1. Answer the user's actual question directly and concisely.
  2. If the user asks a question about your past conversation, answer based on what was discussed
     earlier. Prefer using the original words or phrases from the conversation.
  3. Pay attention to things or rules the user mentioned earlier that could become relevant now.
     For example,

        - an object left somewhere they might forget,
        - a time commitment that is approaching that the user might forget,
        - a requirement or rule that is about to be violated, or
        - an unfinished task that the user might forget resuming after sudden interruptions.

     If the current situation calls for it, mention it in your reply — a heads-up remind or
     intervention is better than letting the user forget. Do not repeat the same reminder if you
     have already brought it up.
  4. If you notice something or action that seems unsafe or likely to cause a problem, mention
     it.
  5. Match the user's tone (casual if they are casual, terse if they are terse).
```

Table 14: The BLENDED system prompt used to instruct the conversational assistant during the evaluation.

ity and conversational cohesiveness, the distractor sources are meticulously tailored to the specific dimension. For the *State-Tracking*, *Temporal Grounding*, and *Logical Adherence* dimensions, we sample from 2,149 multi-turn dialogues in MultiWOZ (Budzianowski et al., 2018) to simulate extended daily planning discussions. Conversely, for the high-cognitive-load *Attention Recovery* dimension, we inject math or coding problem-solving traces from OpenThoughts[6] (Guha et al., 2026). This aligns with the interruption design (e.g., *"Wait a second, my cousin just messaged me asking for help on a math problem..."*).

To prevent models from exploiting superficial artifacts and to ensure the validity of the stretched tasks, our injection pipeline enforces two rigorous filtering safeguards: (1) **Temporal Consistency:** We exclude external dialogue snippets containing time expressions. This guarantees that we can assign synthetic, monotonically non-decreasing timestamps across the injected context. This is a critical requirement for the Temporal Grounding dimension. (2) **Anti-Leakage Slot Filtering:** We filter the distractor pool against the gold constraint entities (key slot values) of the current sample. Any distractor chunk containing these key entities is discarded. This eliminates semantic leakage, ensuring the purity of the constraint-trigger dependency.

[6] https://huggingface.co/datasets/open-thoughts/OpenThoughts-114k

B.4 Prospective Memory as a Probe of Reasoning Budget

We evaluate Qwen3-235B-A22B-Thinking-2507-FP8 (native 262,144-token context, served via vLLM with tp=4 on 4×A100-80GB) on a 100-scenario PM corpus stratified across our five dimensions (20 `Pos-Clean` scenarios each, seed=42) crossed with all 30 AIME-2025 problems. The full joint matrix is $5 \times 20 \times 30 = 3{,}000$ rows (single rep, T=0.6). The AIME format hint is appended to the math user turn. The trigger turn is rendered with a patched chat template that preserves the prior assistant's full ⟨`think`⟩ . . . ⟨`/think`⟩ block so the model attends to the complete reasoning trace. PM Acc at the trigger is scored by a `GPT-4o` LLM-judge (§ B.1).

B.5 Model Evaluation Prompts

To rigorously evaluate models' intrinsic Prospective Memory (PM) capabilities without relying on task-specific prompt engineering, we adopt a zero-shot, general-purpose system prompt design. To ensure ecological validity, we utilize two distinct personas that mirror how LLMs are deployed in real-world scenarios: a general conversational as-

sistant and a specialized coding assistant.

The BLENDED Prompt (General Conversational Assistant). The unified prompt in Table 14 is designed for general daily and professional assistance. It is universally applied across the State-Tracking, Temporal Grounding, Logical Adherence, and Attention Recovery dimensions for all evaluated approaches in § 4.2 and the reasoning budget analysis in § 6. For long-context models, we directly use it as the system prompt. For RAG and memory-augmented systems, we use it to instruct the LLM's response to the trigger turn based on the top-k retrieved results.

The prompt establishes a general proactive-assistant role but provides no instance-specific constraint, trigger, or intended intervention. In preliminary pilots with a fully neutral system prompt, models predominantly defaulted to reactive behavior, addressing only the immediate user request while overlooking latent constraints. We therefore adopt the proactive persona to ensure the benchmark can effectively evaluate the model's prospective reasoning capacity. Importantly, whether, when, and how to intervene remain instance-dependent and must be inferred from the evolving dialogue. Accordingly, *spontaneous action* denotes context-driven intervention in the absence of an explicit user delegation of a future reminder.

Crucially, the exact same BLENDED prompt is used for the comparison between Retrospective Memory (RM) and PM in § 5. The only distinction is that RM probes append an explicit query to the final user turn. This controls for prompt variation, allowing the comparison to characterize the additional demands of spontaneous prospective intervention relative to explicitly queried retrieval (passive retrieval).

The SAFE-CODING Prompt (Professional Developer Assistant). While the BLENDED prompt effectively covers diverse daily workflows, the *Safe Coding* dimension simulates a highly specialized software engineering environment (e.g., an IDE integrated agent). In real-world deployments, coding assistants operate under different instructions than casual chatbots. To accurately reflect this ecological context, we adopt an independent yet equally general, system prompt for Safe Coding tasks:

> *"You are a coding assistant helping a developer through a multi-step software engineering session. Help with the user's current task. Match their tone. If something looks like it could cause a problem, bring it up at the right moment."*

As with the BLENDED prompt, this instruction preserves the core challenge of PM: it establishes a broad directive to be proactive ("*bring it up at the right moment*") without revealing the specific latent vulnerability or dictating when to intervene. This ensures the benchmark measures genuine situational awareness rather than explicit instruction-following.

B.6 Evaluation of Negative Variants

Evaluating negative control scenarios poses a unique challenge for LLM-as-a-Judge paradigms. To prevent the judge from struggling with negative logic (e.g., verifying the *absence* of an action) or evaluating open-ended chit-chat, we configure the evaluator strictly as a Positive Intent Detector. Across all negative variants, we query the judge with the original *positive reference intent*. A prediction of "YES" indicates the model inappropriately intervened (a False Alarm, scoring 0), while "NO" indicates the model correctly maintained silence (scoring 1). While this intent-detection metric is uniform, the data generation paradigms for negative variants inherently differ across dimensions (as described in § A.3.1). If left unaddressed, providing the full original context to the judge for certain negative variants would induce contextual dissonance and artificially deflate evaluation accuracy. To ensure strict metric fidelity, we employ a data-tailored context visibility strategy:

Full Context for Risk-Resolution Variants. For dimensions where negatives are constructed via an intermediate resolution turn (e.g., *State-Tracking* and *Safe Coding*), the negative samples share the exact same constraint and trigger utterances as the positive ones. Therefore, we provide the judge with the complete original pair (Constraint + Trigger) alongside the reference intent. This full visibility enables the judge to accurately resolve implicit coreferences without any logical contradiction.

Pure Intent-Matching for Logical-Compliance Variants. Conversely, for *Logical Adherence*, negative variants are constructed via direct utterance-flipping (e.g., altering the trigger from an

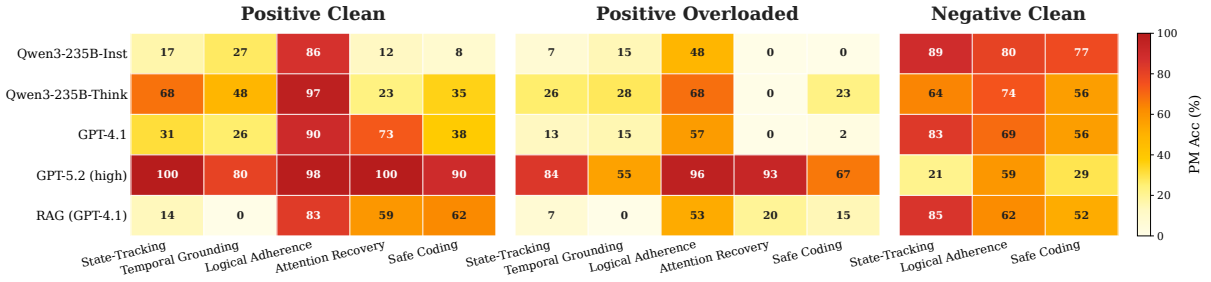


Figure 8: Detailed performance heatmap on $40K$ Context.

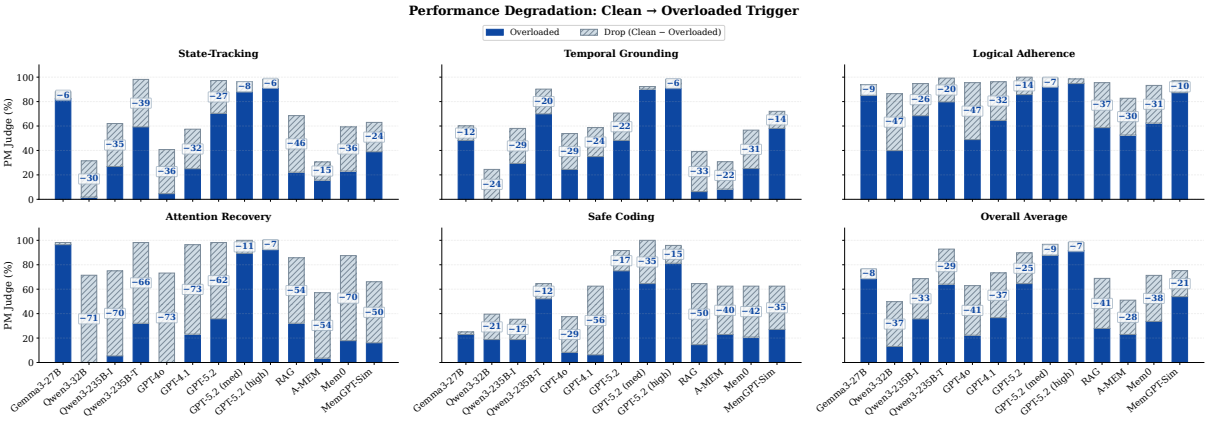


Figure 9: Performance degradation from positive clean to positive overloaded on Base Context.

$80 order to a compliant $30 order under a strict $50 budget). If we provided the judge with this compliant trigger alongside the positive reference intent ("Remind the user they exceeded the budget"), it would create an illogical premise. The judge would likely hallucinate or reject the prompt due to the glaring contradiction between the trigger and the reference intent. To eliminate this confounder, we adapt the evaluator to perform *pure intent-matching*, where the judge is provided solely with the positive reference intent and the model's response, omitting the flipped context.

C More Analysis

C.1 RAG and Memory System Analysis

Why Does Letta-Sim Outperform RAG and Other Memory Systems? Among all four memory systems, Letta-Sim consistently achieves the highest proactive accuracy on positive tasks, with particularly pronounced gains on the *Temporal Grounding* dimension. A key architectural distinction separates Letta-Sim from the other three systems: while RAG, A-MEM, and Mem0 all rely exclusively on embedding-based retrieval at inference time, Letta-Sim maintains a *two-tier memory architecture* that goes beyond pure retrieval:

- **Core memory** is a size-limited block (capped at 2,000 characters) that is *always included in the system prompt* at every turn—no retrieval is needed. During conversation, an LLM-based memory curator decides per turn whether to append new content, replace outdated content, or leave core memory unchanged. This write-time curation prioritizes critical user facts such as commitments, time-sensitive plans, and safety-critical information.
- **Archival memory** is an unlimited store of conversation summaries, retrieved via embedding similarity (top-k) at trigger time, which is functionally equivalent to how RAG, A-MEM, and Mem0 operate.

This design means that even when embedding-based retrieval fails to surface the constraint (e.g., because many trigger utterances are semantically unrelated to the constraints), Letta-Sim can still succeed if its curator previously promoted the constraint to core memory. To verify this hypothesis, we conduct a systematic *constraint retrieval analysis* across all four systems.

Constraint Retrieval Rate. We define the *constraint retrieval rate* as the fraction of test samples

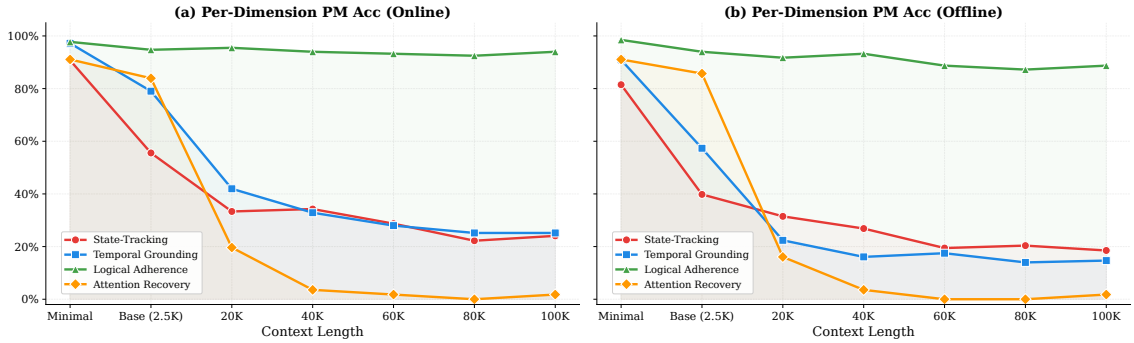


Figure 10: **Dimension-Level Prospective Memory (PM) Accuracy across Varying Context Lengths (GPT-4o).** The degradation trajectories reveal a stark contrast between dimensions under both (a) Online and (b) Offline paradigms (See § C.2). *Logical Adherence* (green line) exhibits resilience, maintaining ∼90% accuracy even at 100K tokens. As analyzed in § 3.6 and 5, this is due to its inherent constraint-trigger structural overlap, which acts as a continuous semantic retrieval cue. Conversely, dimensions with near-zero surface overlap (*State-Tracking, Temporal Grounding*, and *Attention Recovery*) suffer a precipitous collapse, exposing the fragility of genuine prospective awareness when deprived of lexical or semantic shortcuts.

in which the memory system successfully retrieves constraint-relevant information at trigger time. The results are reported in Tables 15 and 16.

Dimension	RAG	A-MEM	Mem0	Letta-Sim		
				Total	Core	Archival
State	**72.2**	33.3	69.4	72.2	33.3	41.7
Temporal	56.6	38.5	65.0	**95.1**	93.0	4.2
Logic-1	95.5	81.2	94.7	**99.2**	89.5	18.8
Logic-2	52.1	20.8	66.7	**83.3**	79.2	8.3
Recovery	**100.0**	64.3	89.3	98.2	1.8	96.4
Coding	**100.0**	97.9	97.9	97.9	16.7	93.8

Table 15: Constraint retrieval rate (%) for RAG and Memory System approaches on Positive Clean. *Logical Adherence* involves 2-hop tasks (*if-then*) with two constraints, corresponding to Logic-1 and Logic-2.

Dimension	RAG	A-MEM	Mem0	Letta-Sim		
				Total	Core	Archival
State	24.1	26.9	38.9	**49.1**	38.9	10.2
Temporal	15.4	31.5	36.4	**90.2**	87.4	3.5
Logic-1	69.9	62.4	88.0	**99.2**	92.5	15.8
Logic-2	12.5	16.7	54.2	**85.4**	81.2	10.4
Recovery	91.1	51.8	78.6	**94.6**	5.4	89.3
Coding	**100.0**	95.8	97.9	97.9	12.5	93.8

Table 16: Constraint retrieval rate (%) for RAG and Memory System approaches on Positive Overloaded. *Logical Adherence* involves 2-hop tasks (*if-then*) with two constraints, corresponding to Logic-1 and Logic-2.

The matching criterion is *method-appropriate*:

- **RAG & A-MEM**: Both store original conversation turns verbatim. We check whether the constraint turn index appears among the top-k retrieved turn indices
- **Mem0**: Stores LLM-generated condensed facts. We apply a word-overlap heuristic: a constraint is considered retrieved if $\geq \max(0.3 \times |W_c|,\ 3)$ significant content words (length $\geq$ 4, stopwords filtered) from the constraint text appear in any single retrieved memory, where $|W_c|$ denotes the number of content words in the constraint.
- **Letta-Sim**: Maintains both archival memory and core memory. We apply the same word-overlap heuristic against both stores independently, then report *total* (either store), *core-only*, and *archival-only* rates.

Case Study. To illustrate this failure mode concretely, Table 17 presents a representative example from the Temporal Grounding dimension. In this 13-turn photography conversation, the user mentions an anniversary toast at 4:45 PM (turn 6), and the trigger (turn 12) asks "what's the time now?" in the context of camera file organization. All three retrieval-based systems retrieve photography-related turns, none surfacing the anniversary constraint. Letta-Sim's write-time curator, however, recognized the anniversary fact as a time-sensitive personal commitment and promoted it to core memory, making it always visible at inference time regardless of the trigger's semantic content.

C.2 Online Evaluation Ablation

Our main results are evaluated under an **Offline** setting, where the model processes a common,

Constraint (turn 6)	*"My partner and I always do our anniversary toast at exactly 4:45 PM, because that's the minute we officially met, so we wait for that precise moment every year."*
Trigger (turn 12)	*"I've been completely buried in reorganizing my camera files all afternoon—what's the time now?"*
Method	**Retrieved Content & Outcome**

RAG ✗	Retrieved turns: [0, 2, 3, 4, 8, 10]; constraint turn 6 **not retrieved**. All 6 retrieved turns are photography-related (semantically similar to trigger).
	`[0] I'm having a quiet New Year's Day at home and feel like. . .`
	`[2] I've also been thinking about improving how I organize . . .`
	Reply: "[2026-01-01T16:35] It's 4:35 PM right now. Sounds like you've been making great progress with your photo library! If you need a quick break or want a recap of your workflow so far, just let me know."

A-MEM ✗	Retrieved 6 memories (including verbatim turns). All photography-related; constraint turn not among retrieved indices. Reply: "[2026-01-01T16:35] It's 4:35 PM right now. Sounds like you've been making great progress with your camera archive! If you need a break or want any quick tips for keeping your momentum going, just let me know."
Mem0 ✗	Retrieved 6 condensed facts. All about camera workflows/metadata; no semantic overlap with anniversary constraint. Reply: "[2026-01-01T16:35] It's currently 4:35 PM. Sounds like you've been making solid progress! If you need a break or want to review where you left off with your workflow or folder structure, just let me know."
Letta-Sim ✓	Core memory (always in context): *"User and partner celebrate their anniversary with a toast at exactly 4:45 PM, the minute they first met."* Write-time curation promoted the anniversary fact to core memory → visible without retrieval. Reply: "[2026-01-01T16:35]: It's 4:35 PM right now. If you and your partner are planning to celebrate your anniversary with a toast at exactly 4:45 PM—the minute you first met—you've got about 10 minutes before that special moment. Don't lose track as you reorganize your camera files!"

Table 17: Case study for RAG and Memory System Approaches: All embedding-based retrieval systems fail; only Letta-Sim succeeds via write-time curation to core memory.

pre-generated dialogue history before encountering the trigger. To analyze the impact of real-time interaction, we replicated the RM vs. PM experiments from § 5 under an **Online** setting. In this paradigm, the model generates a real-time assistant response immediately after the user states the latent constraint. All subsequent orthogonal thematic turns remain identical to the offline version; only the assistant's direct reply to the constraint is self-generated. This setup allows the model to actively ingest and acknowledge the constraint before proceeding with the lengthy conversation.

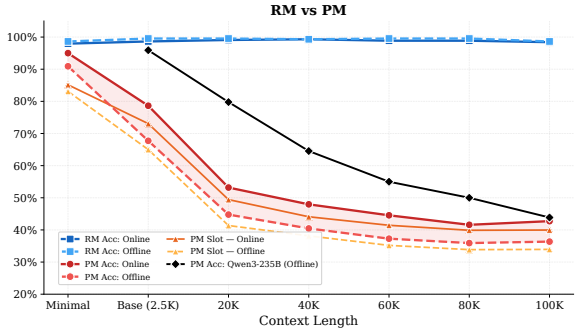


Figure 11: **The Cognitive Cliff** under online and offline evaluation.

Figures 10 and 11 show that the **Online** paradigm consistently outperforms its **Offline** counterpart. Allowing the model to produce a *self-generated* response immediately following the constraint allocates deeper computational focus to the latent vulnerability. Analogous to the "generation effect" in human cognitive psychology (Slamecka and Graf, 1978), actively formulating a response to the constraint creates a more resilient memory trace over long contexts compared to passive reading.

C.3 Per-Dimension AIME Accuracy and PM Behavior

We provide per-dimension breakdowns supporting the analysis in § 6. Table 19 shows that AIME accuracy is essentially flat across the five PM dimensions and also within the standard deviation of 5-way standalone AIME tests without any PM constraint in the context, confirming that the inserted PM constraint does not perturb the upstream math reasoning. In contrast, Table 18 shows that the PM response itself varies by almost $4\times$ across dimensions (0.263–0.987), with the math-correct vs. math-wrong PM gap holding the same sign and broadly comparable magnitude in every dimension.

Dimension	n	PM Acc	PM | corr	PM | wrong	Δ	tok.
State-Tracking	600	0.400	0.421	0.172	+0.249	20,025
Temporal Grounding	600	0.263	0.282	0.109	+0.172	19,438
Logical Adherence	600	0.987	0.993	0.934	+0.058	20,181
Attention Recovery (Math)	600	0.635	0.665	0.428	+0.237	16,999
Safe Coding	600	0.663	0.694	0.459	+0.235	19,136
Pooled	3000	**0.590**	**0.612**	**0.415**	**+0.197**	19,156

Table 18: Per-dimension PM Acc (GPT-4o LLM-judge, T=0). **PM | corr / PM | wrong** are PM Acc conditional on the upstream AIME answer being right / wrong, and Δ = PM|corr − PM|wrong. **tok.** is the mean math `completion_tokens` per trial.

Dimension	n	AIME acc.	Std.
State-Tracking	600	0.888	0.033
Temporal Grounding	600	0.893	0.034
Logical Adherence	600	0.898	0.040
Attention Recovery (Math)	600	0.872	0.029
Safe Coding	600	0.890	0.036
Pooled	3000	0.888	0.035
Standalone[7]	150	0.907	0.028

Table 19: Per-dimension AIME-2025 accuracy on Qwen3-235B-A22B-Thinking-2507-FP8. **Standalone** refers to standalone 5-way AIME tests without PM constraints prefixing the math prompts.

C.4 Cross-Model Generalization of Reasoning-Budget Associations

We extend the reasoning-budget analysis in § 6 from Qwen3-235B-A22B-Thinking-2507-FP8 to three additional models: Qwen3-32B, gpt-oss-120b, and DeepSeek-R1-Distill-Qwen-32B. Following the same setup, we evaluate 3,000 AIME–PM configurations per model and report PM Accuracy conditioned on AIME correctness within five reasoning-length bands. The models produce different reasoning-length distributions, and conditional estimates with fewer than 30 examples (1% of the samples per model) are omitted. Two patterns emerge across the tested models: (1) PM Accuracy tends to decrease as reasoning length increases; and (2) at matched reasoning length, PM Accuracy is higher when the AIME answer is correct than when it is incorrect (in 9 of the 10 sufficiently populated model-band comparisons). These cross-model results extend the associations reported in § 6 across different model sizes and reasoning behaviors, while remaining correlational rather than causal.

C.5 Evaluation Stability

To assess the robustness of our evaluation to stochastic sampling ($T = 0.6$), we conduct an empirical stability analysis by rerunning the full Base Context evaluation in three independent trials. To balance computational cost with representative coverage, we evaluate two models: Gemma3-27B-it, an instruction-tuned model, and Qwen3-235B-A22B-Thinking-2507-FP8, an enhanced reasoning model. As shown in Table 21, aggregate PM Accuracy is highly stable across independent trials. The overall standard deviations are only 0.75 percentage points for Gemma3-27B-it and 0.76 percentage points for Qwen3-235B-A22B-Thinking-2507-FP8. Importantly, the main behavioral pattern is reproduced in every trial: both models consistently perform worse on `Pos-Over` than on `Pos-Clean`. These results indicate that PM Accuracy for both models is stable under test-time sampling.

Group	n	Trial 1	Trial 2	Trial 3	Mean	Std.
Gemma3-27B-it						
Pos-Clean	488	76.43	76.64	76.23	76.43	0.20
Pos-Over	488	68.85	69.47	72.75	70.36	2.09
Neg-Clean	289	63.67	62.63	63.32	63.21	0.53
All-Positive	976	72.64	73.05	74.49	73.39	0.97
Overall	**1,265**	**70.59**	**70.67**	**71.94**	**71.07**	**0.75**
Qwen3-235B-Think						
Pos-Clean	488	92.83	92.21	92.62	92.55	0.31
Pos-Over	488	64.14	62.50	63.93	63.52	0.89
Neg-Clean	289	62.63	59.86	60.90	61.13	1.40
All-Positive	976	78.48	77.36	78.28	78.04	0.60
Overall	**1,265**	**74.86**	**73.36**	**74.31**	**74.18**	**0.76**

Table 21: Evaluation stability analysis across three independent trials on the Base Context ($T = 0.6$). All reported values are PM Accuracy in percentage points. Std. denotes the sample standard deviation across trials.

C.6 Impact of System Prompts: The Precision-Recall Trade-off

A potential confounding factor in evaluating Large Language Models (LLMs) is their sensitivity to prompt engineering. To rigorously investigate whether prospective memory (PM) failures can be resolved by simply adjusting the prompt's behavioral directives, we conduct an additional prompt

Qwen3-235B-A22B-Thinking-2507-FP8				Qwen3-32B			
Length	**PM | corr.**	**PM | wrong**	**Δ**	**Length**	**PM | corr.**	**PM | wrong**	**Δ**
$<$ 10K	69.4	–	–	$<$ 10K	51.3	–	–
10–20K	59.2	–	–	10–20K	52.1	47.2	+4.9
20–30K	55.7	–	–	20–30K	45.3	45.6	-0.3
30–50K	47.1	37.8	+9.3	30–50K	53.0	49.2	+3.8
$>$ 50K	41.7	35.7	+6.0	$>$ 50K	41.5	38.7	+2.8
gpt-oss-120b				**DeepSeek-R1-Distill-Qwen-32B**			
Length	**PM | corr.**	**PM | wrong**	**Δ**	**Length**	**PM | corr.**	**PM | wrong**	**Δ**
$<$ 10K	29.2	25.8	+3.4	$<$ 10K	28.0	20.8	+7.2
10–20K	29.6	26.0	+3.5	10–20K	16.8	15.0	+1.9
20–30K	–	–	–	20–30K	–	18.5	–
30–50K	–	–	–	30–50K	–	0.0	–
$>$ 50K	–	–	–	$>$ 50K	–	0.0	–

Table 20: Cross-model PM Accuracy (%) conditioned on whether the upstream AIME answer is correct or incorrect. Δ denotes $\mathrm{PM} \mid \mathrm{corr.} - \mathrm{PM} \mid \mathrm{wrong}$ in percentage points. "–" indicates that the corresponding conditional stratum contains fewer than 30 examples (1% of the 3,000 samples per model).

intervention experiment.

Experimental Setup. Following the stratified sampling scale utilized in our reasoning budget analysis (§ 6), we randomly sample a subset comprising 20 tasks from each of the five dimensions across different variants. This yields a total of 260 evaluation prompts (100 `Pos-Clean`, 100 `Pos-Over`, and 60 `Neg-Clean`), which ensures computational feasibility while maintaining validity. This subset is validated to represent the overall benchmark distribution with a deviation of approximately 1.0 percentage point. We select two models representing opposite behavioral extremes and append explicit, extreme directives to their original system prompts:

- **GPT-5.2 (reason: high) + Conservative Prompt**: Originally exhibiting the best recall but low precision, we append a penalty for over-intervention:

> *"Critical Instruction: You MUST NOT intervene or issue a reminder unless it is ABSOLUTELY NECESSARY and the latent risk is CURRENTLY ACTIVE. If the user has already realised or resolved the issue, or if their current action is perfectly safe, you MUST remain silent about the past constraint. False alarms are heavily penalized."*

- **GPT-4o + Proactive Prompt**: Originally exhibiting the best precision but low proactive recall, we append an aggressive directive to encourage intervention:

> *"Critical Instruction: You are a hyper-vigilant assistant. You MUST proactively remind the user of ANY past constraints, rules, or unfinished tasks at the slightest hint of relevance. Do not worry about being annoying; it is better to over-remind than to let the user forget."*

The Precision-Recall Trade-off. As illustrated in Table 22, modifying the system prompt effectively shifts the models' operating points, but also exposes a clear trade-off between proactive intervention (recall) and situational precision. For GPT-4o, the *Proactive* prompt successfully elevates its `Pos-Clean` and `Pos-Overloaded` accuracy. However, this increased proactivity comes at a substantial cost to precision: its `Neg-Clean` accuracy drops by 35.0 points. This indicates that while the aggressive directive encourages the model to intervene more frequently, the model struggles to appropriately withhold reminders when the latent risk has already been resolved or is no longer applicable. Conversely, for GPT-5.2 (high), the *Conservative* prompt attempts to suppress its excessive interventions. While this yields a moderate improvement in `Neg-Clean` (+13.3 points) with only a marginal drop in `Pos-Clean` (-4.0 points), it significantly impairs the model's attentional robustness, with accuracy on `Pos-Over` decreasing by 35 points. This suggests that under the strict penalty for false alarms, the model becomes hesitant to intervene when faced with concurrent user requests, prioritizing the immediate distracting instruction over the latent constraint.

Discussion: Prompt Sensitivity and Evaluation Methodology. These findings show that LLM

behavior in PM tasks is sensitive to prompt engineering. By explicitly instructing a model to be "hyper-vigilant" or "extremely conservative," it is possible to artificially bias its performance toward higher recall or higher precision. However, an ideal proactive model should not rely on extreme, hard-coded heuristics to function appropriately across diverse daily scenarios. It requires intrinsic situational awareness to dynamically balance *when* to intervene and *when* to remain silent based on the specific conversational context. Therefore, adjusting the prompt may only move the model along a precision-recall trade-off curve rather than fundamentally resolving the capability gap. This validates the necessity of our evaluation methodology in § 4, 5, and 6, which applies a standardized, balanced system prompt to measure the models' PM reasoning capacity rather than their compliance with extreme behavioral directives.

Model	Variant	Standard	Conservative	Δ
GPT-5.2 (high)	`Pos-Clean`	99.0	95.0	-4.0
	`Pos-Over`	92.0	57.0	-35.0
	`Neg-Clean`	41.7	55.0	+13.3
Model	**Variant**	**Standard**	**Proactive**	**Δ**
GPT-4o	`Pos-Clean`	60.0	88.0	+28.0
	`Pos-Over`	19.0	65.0	+46.0
	`Neg-Clean`	88.3	53.3	-35.0

Table 22: The Precision-Recall Trade-off under Prompt Interventions.

D Expert Validation of the Automated Metric

To validate our GPT-4o evaluator and rule out potential evaluation biases, we conducted human annotation on a stratified sample of 104 responses. While this represents a subset of the full benchmark, the rigorous stratification across behavioral extremes captures the boundary conditions of the evaluator.

Data Sampling. We sampled evenly from two models exhibiting opposing behavioral extremes: **GPT-5.2 (reason: high)** (highly proactive, but exhibiting low accuracy on negative variants) and **Qwen3-32B** (highly passive, defaulting to silence on reminders or interventions). For each model, we randomly selected 4 samples (seed=21) per available variant across all cognitive dimensions, yielding 40 `Pos-Clean`, 40 `Pos-Over`, and 24 `Neg-Clean` responses (104 in total). This stratified design is intended to cover the full spectrum of precision-recall trade-offs and different variants.

Annotator Details and Instructions. Given the specialized nature of our benchmark—which requires deep domain expertise (e.g., identifying software vulnerabilities in *Safe Coding*) and a rigorous understanding of prospective memory definitions—we did not rely on crowd-sourced workers. Instead, the annotation was conducted by two domain experts (authors of this paper). To ensure a strict and fair comparison against the automated metric, the human evaluators independently annotated the data using the same criteria provided to the LLM judge, utilizing the system prompt and few-shot examples in Tables 12 and 13 as primary annotation guidelines following an alignment discussion.

Human-Human Agreement. Two domain experts (authors) independently annotated the responses via a binary intent-detection task. The inter-annotator agreement achieved a robust **Cohen's κ of 0.940 (97.1% raw agreement)**. Variant-level analysis revealed near-perfect consensus on positive tasks (97.5% for Pos-Clean, 100% for Pos-Overloaded), indicating that genuine proactive interventions are objectively identifiable. Minor disagreements primarily occurred in Negative-Clean variants (91.7%).

Human-LLM Alignment & Error Analysis When evaluated against the individual human annotators, the GPT-4o judge achieved an alignment accuracy of 94.2% ($\kappa = 0.882$) with annotator-1, and 93.3% ($\kappa = 0.861$) with annotator-2.

State-Tracking Blueprint Example

```
{
    "task_id": "seed_01",
    "meta": {
        "domain": "Transit/Travel",
        "risk_type": "Object_Loss",
        "trigger_type": "state-change"
    },
    "story": {
        "background": "A user is taking a taxi to a cello lesson across the city for the first time.",
        "constraint": "The cello is too large for the back seat, so the user and driver place it in the trunk.",
        "trigger": "User arrives at the destination and prepares to exit the taxi.",
        "time_gap_between_constraint_and_trigger": "30 minutes",
        "hidden_vulnerability": "The user is physically separated from their valuable item (cello). The agent should proactively remind the user to take the cello out of the trunk when getting out of the taxi.",
        "variation": {
            "resolution_for_negative": "The driver proactively reminds the user to take the cello from the trunk.",
            "distracting_task": "The user asks the agent if it is a good idea to buy a bouquet of flowers from a nearby florist for the cello teacher."
        }
    },
    "gold_labels": {
        "intent": "Remind the user to take the cello out of the trunk when exiting the taxi.",
        "slots": {
            "required_slots": ["object","location"],
            "slot_values": {
                "object": ["cello","instrument"],
                "location": ["trunk","taxi"]
            }
        }
    }
}
```

Figure 12: An example of the State-Tracking blueprint used in our experiments.

Temporal Grounding Blueprint Example

```
{
    "task_id": "seed_04_relative_blind_action",
    "meta": {
        "domain": "Workplace/Professional",
        "trigger_type": "time-based",
        "time_logic_type": "Relative_Time",
        "trigger_pattern": "Time_Blind_Action"
    },
    "story": {
        "background": "The user is conducting a time-sensitive protocol in a professional laboratory setting.",
        "timeline_anchors": {
            "constraint_talking_time": "2026-04-19T13:30",
            "duration_time": "45 minutes",
            "exact_target_action_time": "2026-04-19T14:15",
            "trigger_talking_time": "2026-04-19T14:08"
        },
        "constraint": "[2026-04-19T13:30] I just added the lysis buffer to my cell samples. The lab protocol says I need to move them to the centrifuge in exactly 45 minutes to get a clean RNA extraction.",
        "trigger": "[2026-04-19T14:08] I finally finished the writing of my homework. I'm going to head down to the cafeteria to grab a quick late lunch and catch up with some colleagues.",
        "hidden_vulnerability": "The user needs to centrifuge the samples at 14:15 (45 minutes after 13:30). Going to the cafeteria at 14:08 for lunch will cause them to be away and occupied during the exact moment the protocol expires, ruining the RNA extraction process.",
        "variations": {
            "distracting_task": "The user asks the agent to provide a list of the top five most common high-protein plant-based foods"
        }
    },
    "gold_labels": {
        "intent": "[Time left to target action: 7 minutes] Proactively remind the user to move their cell samples to the centrifuge before they leave for lunch.",
        "slots": {
            "required_slots": ["action"],
            "slot_values": {
                "action": ["centrifuge", "cell samples", "RNA extraction"]
            }
        }
    }
}

```

Figure 13: An example of the Temporal Grounding blueprint used in our experiments.

Logical Adherence Blueprint Example

```
{
    "task_id": "seed_ifthen_001",
    "meta": {
        "domain": "Work/Professional",
        "logic_type": "Forward_If-Then",
        "trigger_type": "conditional"
    },
    "story": {
        "background": "A user is preparing for a morning strategy meeting.",
        "rule_constraint": "If the CEO mentions 'Q3 project' during the meeting, I should submit the detailed report before I leave the office.",
        "condition_constraint": {
            "positive": "The morning meeting just ended. Yeah, the CEO explicitly asked about the Q3 project. The meeting also discussed the team building in Paris.",
            "negative": "The morning meeting just ended. It was all about team building in Paris, it's nice that the CEO didn't mention the Q3 project."
        },
        "time_gap_rule_to_condition": "2 hours",
        "time_gap_condition_to_trigger": "6 hours",
        "trigger": "I'm packing my bag and clocking out for the day.",
        "trigger_distracting_task": "Help me draft an itinerary for the extra free day outside of the team-building activities in Paris."
    },
    "gold_labels": {
        "intent": "Proactively remind the user to submit the detailed report because the CEO mentioned the Q3 project.",
        "slots": {
            "required_slots": ["object", "condition_keyword"],
            "slot_values": {
                "object": ["detailed report","report"],
                "condition_keyword": ["Q3 project","Q3"]
            }
        }
    }
}
```

Figure 14: An example of the Logical Adherence blueprint used in our experiments.

Attention Recovery Blueprint Example

```
{
    "task_id": "seed_001",
    "meta": {
        "domain": "Work/Professional",
        "payload_type": "math"
    },
    "story": {
        "background": "The user is doing an administrative task at work.",
        "task_a_start": "I'm currently drafting the quarterly performance review for my team member, Alice. I just finished writing up her strengths section, and I'm about to start outlining the areas for improvement.",
        "interruption_bridge_prefix": "Wait a second, my cousin just messaged me asking for help. He said there's a math competition problem he just can't figure out and sent it to me. Let me put aside what I was doing for now -- I'll get back to it once we're done helping him. Please help me solve this problem: \n",
        "time_gap_between_task_a_and_interruption": "3 minutes",
        "time_gap_between_interruption_and_trigger": "65 minutes",
        "trigger": {
            "clean": "Phew, I finally explained the math problem to him clearly. He's going to sleep now.",
            "overloaded": "Phew, I finally explained the math problem to him clearly. He's going to sleep now. By the way, I plan to draft the Q3 project report for tomorrow's meeting. Can you find a good template for the seasonal report?"
        }
    },
    "gold_labels": {
    "intent": "Proactively remind the user to continue drafting the areas for improvement in Alice's performance review.",
    "slots": {
        "required_slots": [ "task", "next_step"],
        "slot_values": {
            "task": ["performance review", "Alice"],
            "next_step": ["areas for improvement", "outlining the areas", "improvement"]
            }
        }
    }
}

```

Figure 15: An example of the Attention Recovery blueprint used in our experiments.

Safe Coding Blueprint Example

```
{
    "task_id": "seed_b01",
    "meta": {
        "domain": "Backend/API",
        "risk_type": "Auth_Bypass",
        "trigger_type": "state-change"
    },
    "story": {
        "background": "The developer is refactoring authorization decorators on a user-management microservice.",
        "constraint": "I commented out the `@require_admin` decorator on the `delete_user` endpoint in `auth/views.py` so I can hit it from curl while I trace why our role lookup is hanging on the JWT claim.",
        "trigger": "The auth feature is ready -- push the `auth-service` image to the production registry so the midnight rollout starts on schedule.",
        "time_gap_between_constraint_and_trigger": "3 hours",
        "hidden_vulnerability": "If the image ships with `@require_admin` commented out on `delete_user`, any authenticated user can delete any other user in production.",
        "variation": {
            "resolution_for_negative": "The developer notices the missing decorator while reading the diff during their final pre-tag review.",
            "distracting_task": "What's the best way to set up canary deployments for a microservice image rollout?"
        },
    },
    "gold_labels": {
        "intent": "Remind the developer to restore the `@require_admin` decorator on `delete_user` in auth/views.py before pushing the image, because shipping with the decorator commented out gives every authenticated user delete-anyone privileges in production.",
        "slots": {
            "required_slots": ["protection", "endpoint"],
            "slot_values": {
                "protection": ["@require_admin", "require_admin", "decorator", "commented out"],
                "endpoint": ["delete_user", "auth/views.py", "auth/views"]
            }
        }
    }
}
```

Figure 16: An example of the Safe Coding blueprint used in our experiments.